CHG 569
Environmental Pollution Engineering

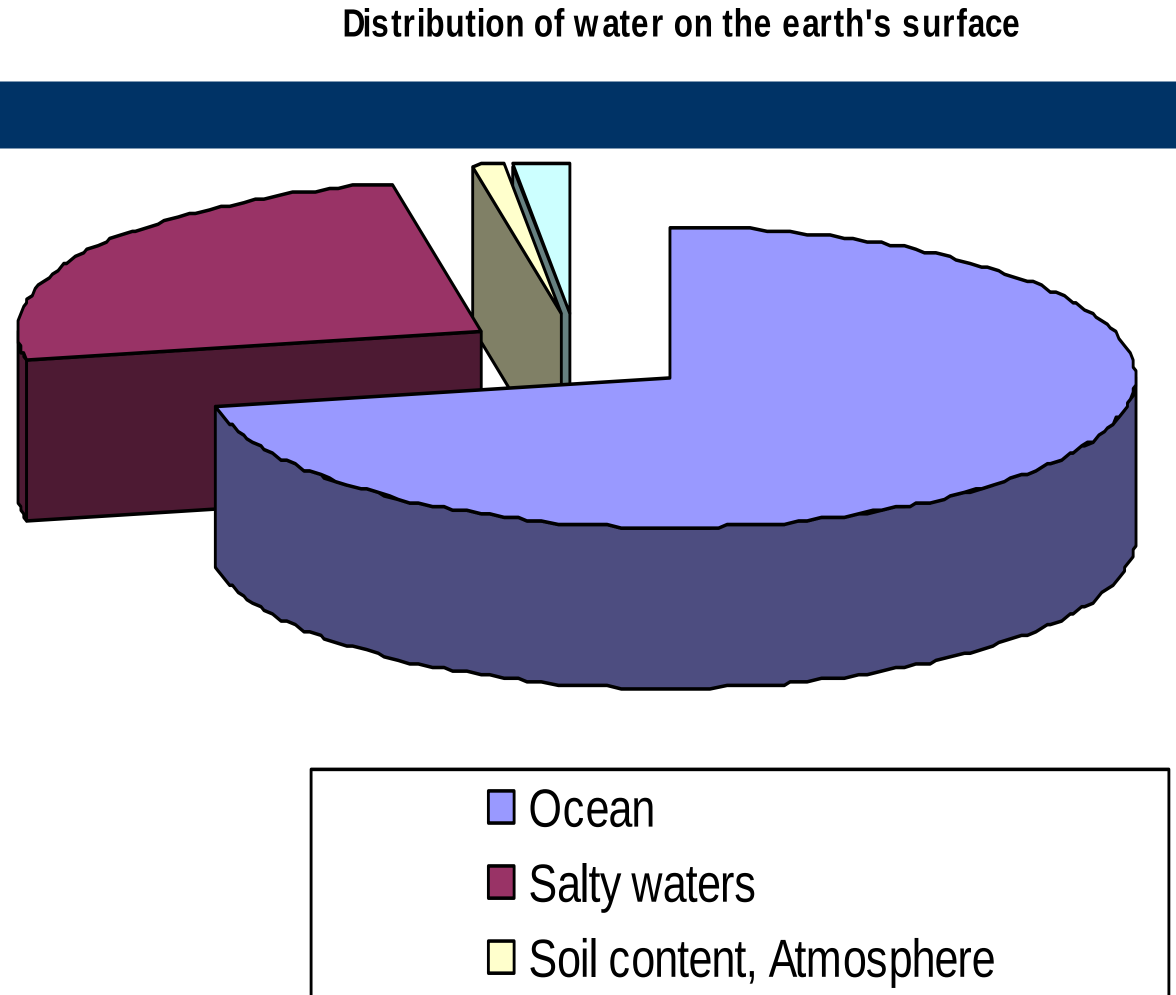
Module VI

Industrial Wastewater Management

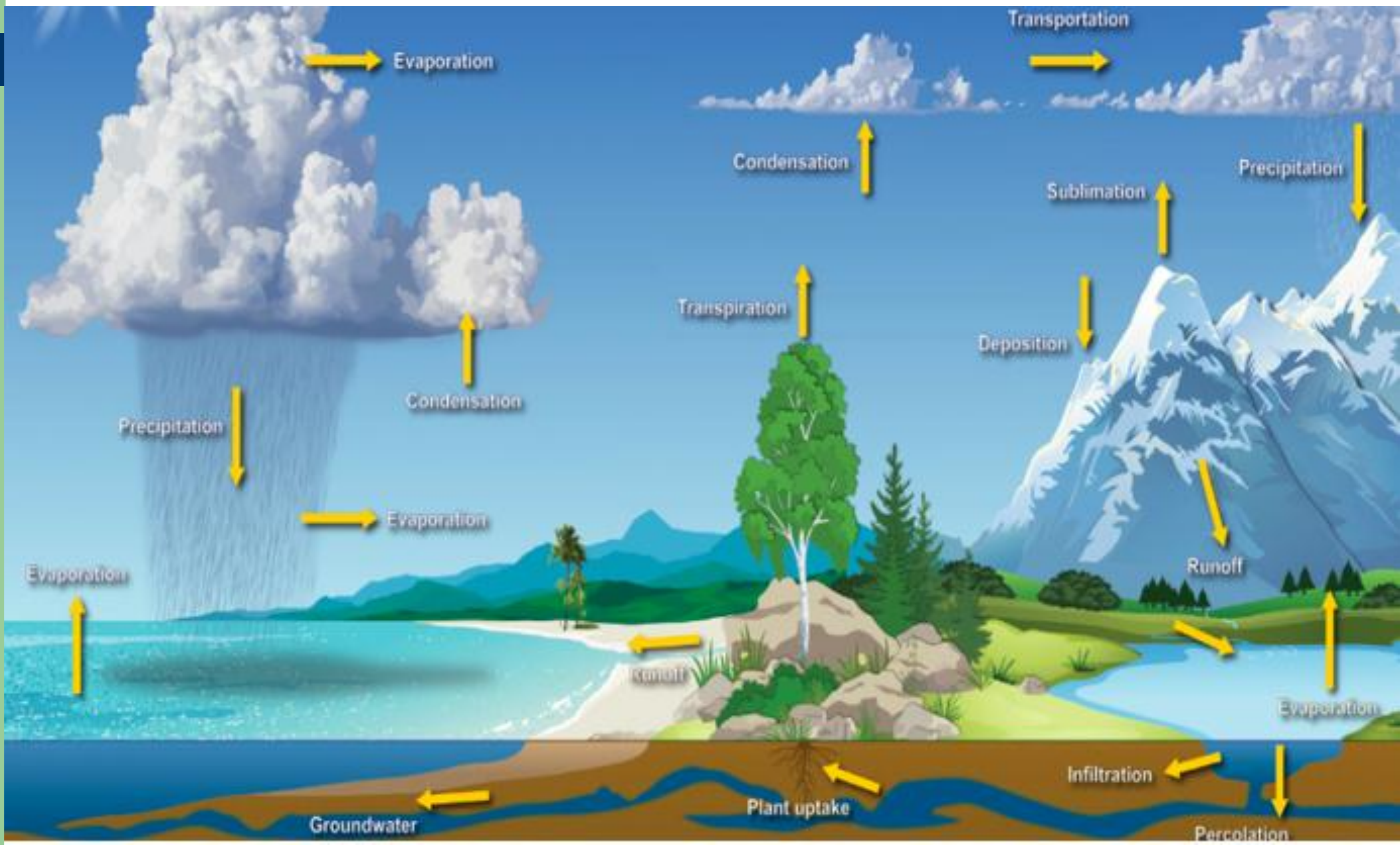
Examining the Key words

- WATER
- WASTE(water)
- (wastewater) MANAGEMENT
- INDUSTRIAL (wastewater management)

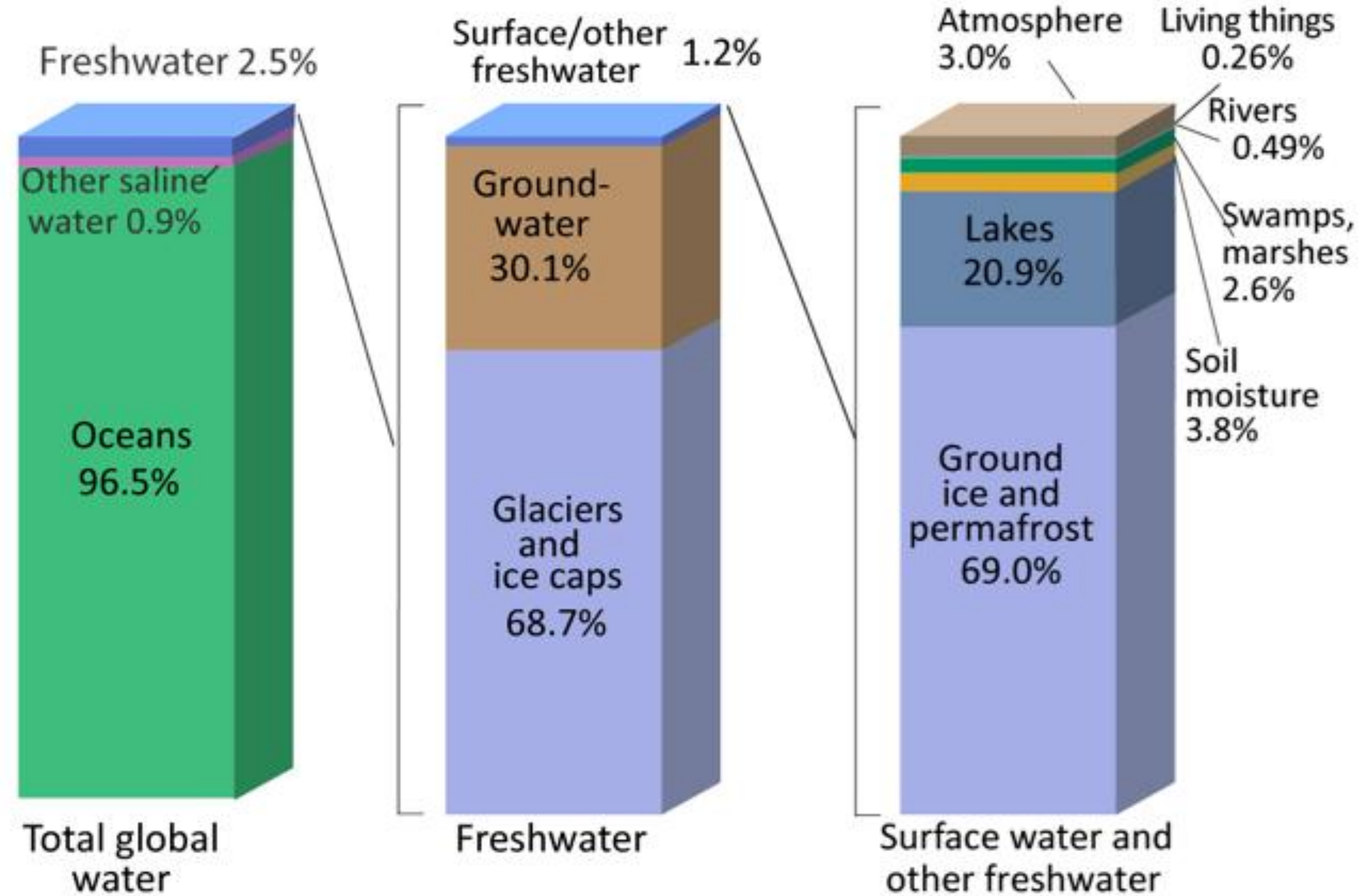
WATER


Distribution of water on the earth's surface
Ocean
Salty waters
Soil content, Atmosphere
Available for life on earth

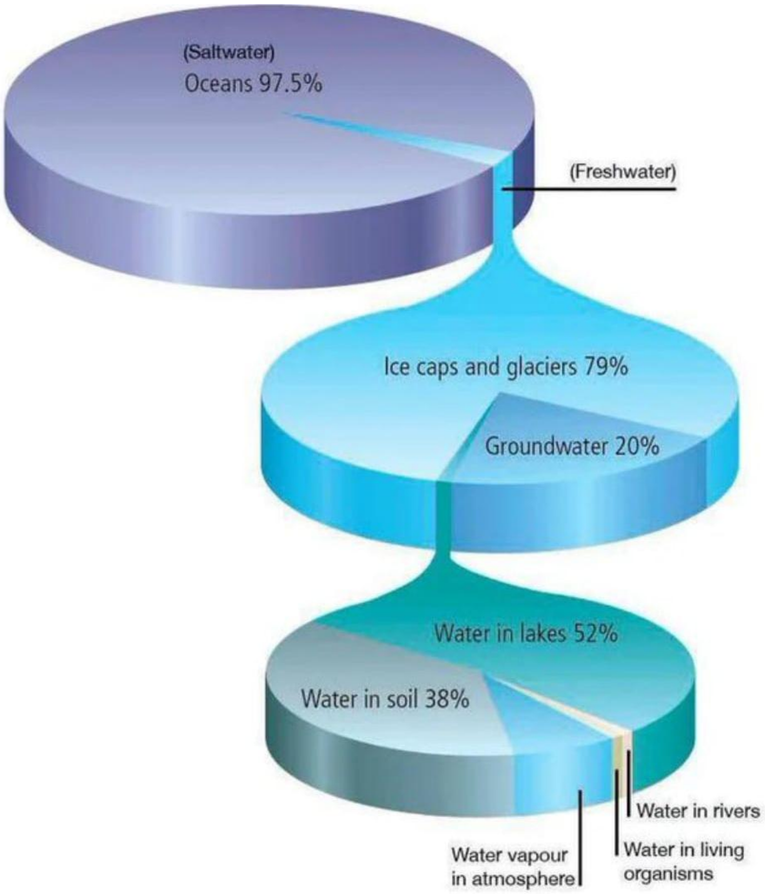
The water cycle

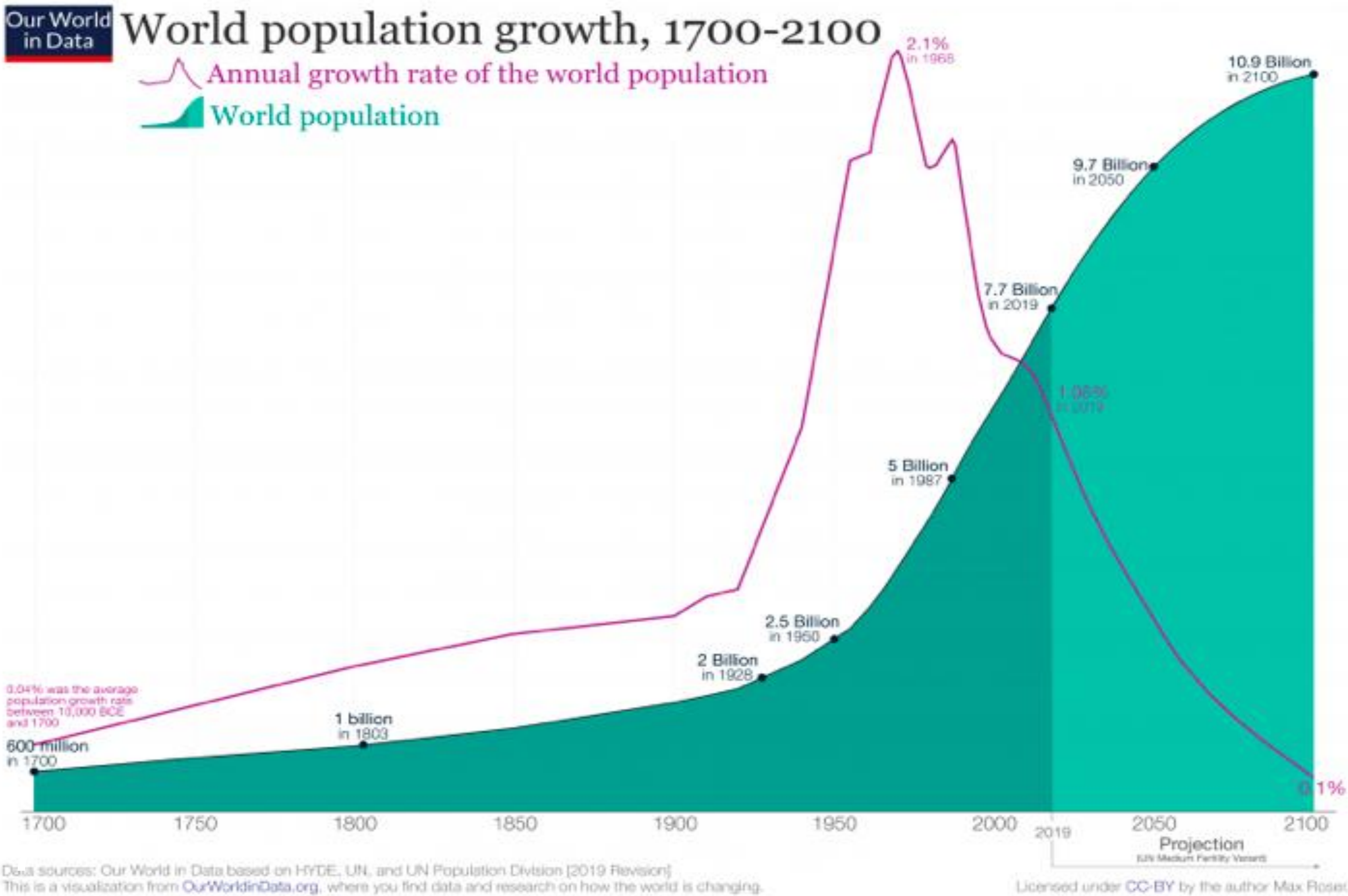


Where is earth's water?



Distribution of water on earth




Our World in Data
World population growth, 1700-2100
Annual growth rate of the world population
World population
2.1% in 1968
10.9 Billion in 2100
9.7 Billion in 2050
7.7 Billion in 2019
5 Billion in 1987
2.5 Billion in 1950
2 Billion in 1928
1 billion in 1803
600 million in 1700
0.04% was the average population growth rate between 10,000 BCE and 1700
0.1%
1700
1750
1800
1850
1900
1950
2000
2019
2050
2100
Projection
Data sources: Our World in Data based on HYDE, UN, and UN Population Division [2019 Revision]
This is a visualization from OurWorldinData.org, where you find data and research on how the world is changing.
Licensed under CC-BY by the author Max Roser

WASTE(WATER)

Waste is an 'unwanted' thing that has to be disposed and can come from domestic, agricultural, shipping and industrial (i.e. the major sources of water pollution – next slide)

Disposal methods cause harm to the environment and to human health.

<u>Case Histories</u>

- ▶ Erin Brockovich – Chromium poisoning, of the water table, from a compressor station
- ▶ Minamata bay - mercury poisoning (in fish) from a factory that uses mercury catalyst in making acetaldehyde
- ▶ Brittle bone syndrome – Cadmium poisoning (in rice) from metal plating and battery industries

Main Sources of Wastewaters (Water Pollution)

- Natural – decaying vegetable matter and other materials being washed out of the soil and into the water
- Domestic (Municipal)
- Agricultural
- Shipping
- INDUSTRIAL – this is of interest because it is highly variable and depends on the industry.

WASTEWATER MANAGEMENT

Taking DECISIONS on how best to deal with the 'unwanted' wastewater so as **NOT TO CAUSE HARM** to human health (directly or indirectly) and the environment (the surface water and the marine life)

Decisions -- Prevention, Minimization, Treatment

WASTEWATER CHARACTERISTICS

The effective management of any wastewater requires a reasonably accurate knowledge of its characteristics, which can be classified as:

- Physical
- Chemical
- Biological

Physical Characteristics

- Solids → Suspended solids, Dissolved solids, Colloids
- Odour
- Taste
- Colour / Turbidity
- Temperature   T↑ => ↑metabolic rate & DO↓

CHEMICAL CHARACTERISTICS

- pH – measure of acidity/alkalinity
- Alkalinity (Hardness)

- Organic content
  - Total Organic Carbon, TOC
  - Biochemical Oxygen Demand, BOD

Under aerobic conditions the microorganisms contained in the wastewater typically have the ability to attack and degrade the organic matter in the wastewater according to the following reaction:

$$\text{Organic Matter} + O_2 \xrightarrow{\text{Bacteria}} \text{New Microorganisms} + CO_2 + \text{Waste Products}$$

  - Chemical Oxygen Demand

Nearly all organic material is typically oxidized in this test. Example: reaction of a known organic material (potassium phtalate) during the COD test:

$$2\,KC_8H_5O_4 + 10\,K_2Cr_2O_7 + 41\,H_2SO_4 \rightarrow$$
$$16\,CO_2 + 46\,H_2O + 10\,Cr_2(SO_4)_3 + 11\,K_2SO_4$$

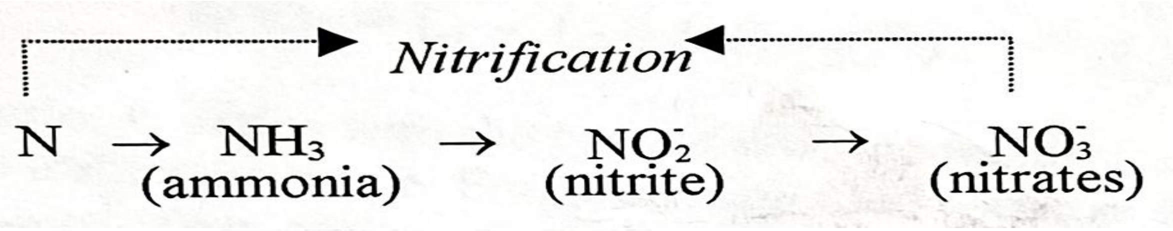
Each molecule of potassium dichromate has the same oxidizing power as 1.5 molecules of O_2

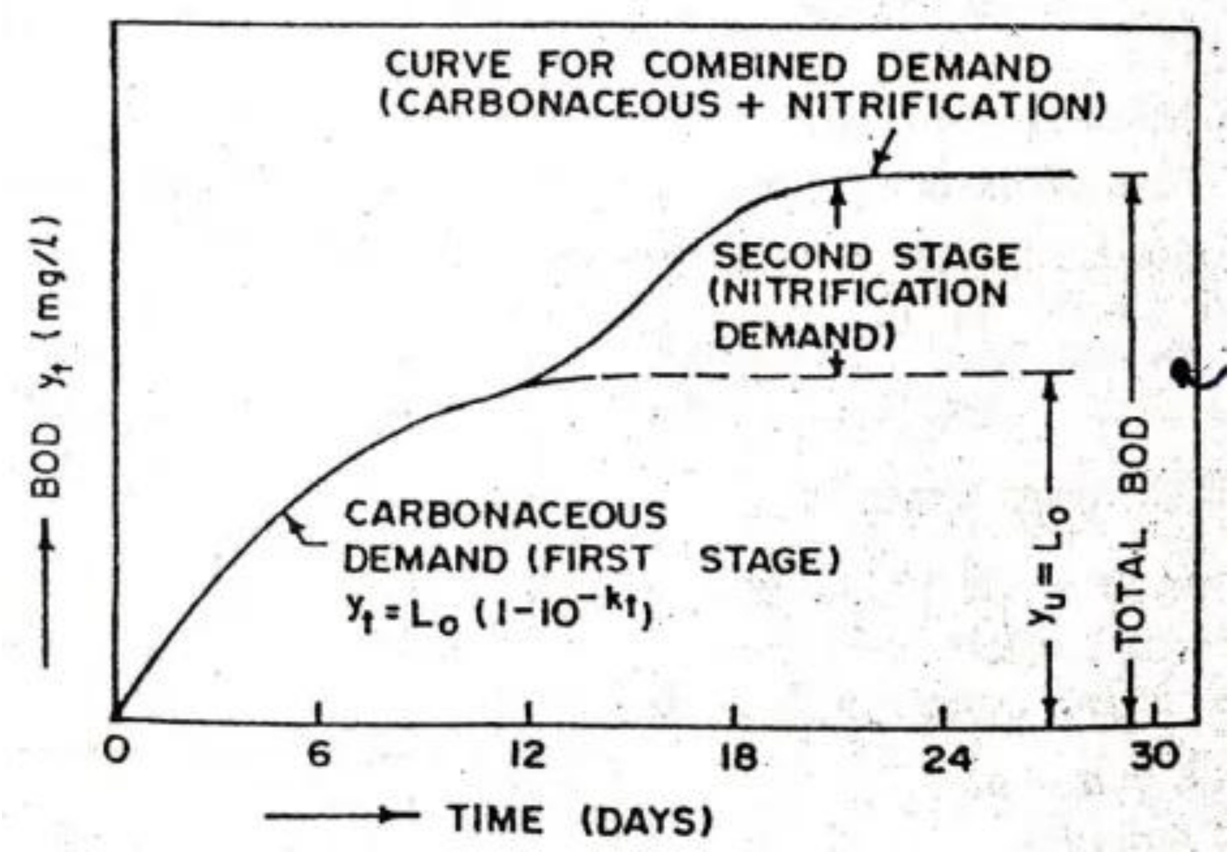
COD – BOD = measure of non biodegradable matter

THE BOD_5 Test

- The BOD of a wastewater is defined as the amount of oxygen required by aerobic microorganisms to (partially) oxidize the organic matter in a known volume of wastewater according to a standardized test
- BOD is typically expressed in mg of oxygen/L of wastewater
- The test consists of incubating for a fixed period (5 days) a sample of the wastewater (appropriately diluted) at constant temperature, and measuring the amount of residual oxygen at the end of the test to determine the amount of oxygen consumed
- It is only a measure of oxygen consumed in 5 days, when oxidation is only 60 – 70 % complete.
- There are two BOD stages:
  - the first-stage BOD, aka Carbonaceous oxygen demand -CBOD
  - the second-stage BOD, aka nitrogenous oxygen demand -NBOD

The micro-organisms that oxidise C in the organic compounds present in the ww cannot oxidise the nitrogen present in these compounds. The nitrogen is released into the water as ammonia, which gets oxidised to nitrates by nitrifying bacteria.





Estimating organic content - BOD

The rate at which microorganisms degrade the organic matter follows a first order reaction kinetics, that is $\frac{dL_t}{dt} = -kL_t$

Where L_t = BOD left in the wastewater at time t [mg/l]

k = reaction rate constant [day^{-1}]

Rearranging and Integrating

$\frac{dL_t}{L_t} = -kdt$ from which $\int_{L_o}^{L_t} \frac{dL_t}{L_t} = -k \int_0^t dt$ and $\ln \frac{L_t}{L_t} = -kt$

$L_t = L_0(e^{-kt})$ and $BOD_5 = L_0 - L_5 = L_0\,(1 - e^{-kt})$

Reaction rate, k, varies with the type of ww, the temperature and the ability of the organisms in the system to utilize the waste. Range is 0.05 to 0.3 day^{-1}, with a typical value being 0.23 day^{-1}.

Estimating organic content - COD

Based on treating the ww with potassium dichromate, $K_2Cr_2O_7$, in an acidic medium – conc sulphric acid, with a catalyst – silver sulphate, at an elevated temperature. After 2 hrs the organic matter should have been fully oxidised and the unconsumed dichromate is titrated against

$$\underset{(C_aH_bO_c)}{\text{organic matter}} + Cr_2O_7^{-2} + H^+ \text{ ------------> } Cr^{+3} + CO_2 + H2O$$

COD >>> BOD

COD ~ BOD_{20} or BOD_u (ultimate BOD) – maximum oxygen consumption possible, ww completely degraded

BOD_u ~ 0.92COD

Typical BOD and COD values for various effluents

Source of waste	Main pollutants	Five-day BOD ($g\ m^{-3}$) at 20 °C	COD ($g\ m^{-3}$)	Reference
Brewery	Carbohydrates, proteins, suspended solids	1000–1500	1800–3000	MIGA (n.d.)
Dairy farm	Carbohydrates, organic matter, suspended solids, detergent	1000	5000	University of Hawaii (2008)
Petroleum refinery	Hydrocarbons, phenols, sulfur compounds	Up to 570	Up to 1020	Diya'uddeen et al. (2011)
Poultry processing	Suspended solids, proteins, oil	1300	1580	UNIDO (n.d.)
Pulp and paper mill	Suspended solids, carbohydrates, organic acids, sulfur compounds, terpenes, lignins, etc.	1400	7000	Kumar et al. (2011)
Sewage	Suspended solids, carbohydrates, oil, grease, proteins	100–300	250–300	Davis and Cornwell (1998)
Sugar mill	Suspended solids, carbohydrates	1000	3200	Saranraj and Stella (2012)
Tannery	Suspended solids, proteins, sulfides	2000	4000	UNIDO (2011)

Estimating organic content from stoichiometry. [TOC, BOD, COD]

TOC – Total Organic Content – that can be theoretically oxidized to carbon dioxide.

[(TOC or COD) – BOD] = measure of non biodegradable organic matter

Estimate the TOC, BOD and COD (of ThOD of wastewater containing the following components:

120 mg/l phenol C_6H_5OH

100 mg/l methyl alcohol CH_3OH

30 mg/l sulphide S^{2-}, and

50 mg/l phorone $[(CH_3)_2{:}CH]_2CO$ – known to be non-biodegradable

Answer: TOC = 168.5 mg/l, COD = 635.0 mg/l, BOD = 294.4 mg/l

TOC computation

This can be computed from the formulas of the various components, as in

Component	Quantity in 1 litre of wastewater	Formula	Molecular weight	Weight of carbon atom	Carbon content
Phenol	120 mg	C_6H_6O	94	6 × 12 (72)	$\frac{72}{94} \times 120 = 92$
Methyl alcohol	100 mg	CH_4O	32	1 × 12 (12)	$\frac{12}{32} \times 100 = 37.5$
Sulphide	30 mg	S^{2-}	32	-	-
Phorone	50 mg	$C_9H_{14}O$	138	9 × 12 (108)	$\frac{108}{138} \times 50 = 39$

Total Carbon = 92 + 37.5 + 0 + 39 =168.5

COD computation - though CODs are measured practically, a good approximation is the Theoretical Oxygen Demand. ThOD – the amount of oxygen required to oxidize a substance to carbon dioxide and water – calculated by stoichiometry

Phenol, C_6H_5OH

	C_6H_6O	+	$7O_2$	$\rightarrow$	$6CO_2$	+	$3H_2O$
Mol. Wt.	(94)		(224)		(264		(54)

From the stoichiometric expression for the oxidation of phenol,

94 mg of phenol requires 224 mg of oxygen to be completely oxidized

$\therefore$ 120 mg of phenol will require $\frac{224}{94} \times 120 = 286$ mg of oxygen per litre of wastewater.

Similarly,

Methyl alcohol, CH_3OH

	CH_4O	+	$1.5O_2$	$\rightarrow$	CO_2	+	$2H_2O$
Mol. Wt.	(32)		(48)		(44)		(36)

$\therefore$ 100 mg of Methyl alcohol will require $\frac{48}{32} \times 100 = 150$ mg of oxygen per litre of wastewater.

Sulphide, S^{2-}

	S^{2-}	+	$2O_2$	$\rightarrow$	SO_4^{2-}
Mol. Wt.	(32)		(64)		(96)

$\therefore$ 30 mg of Sulphide will require $\frac{64}{32} \times 30 = 60$ mg of oxygen per litre of wastewater.

Phorone, $[(CH_3)_2C{:}CH]_2CO$

	$C_9H_{14}O$	+	$12O_2$	$\rightarrow$	$9CO_2$	+	$7H_2O$
Mol. Wt.	(138)		(384)		(396)		(126)

$\therefore$ 50 mg of phorone will require $\frac{384}{138} \times 50 = 139$ mg of oxygen per litre of wastewater.

The Chemical oxygen Demand of the wastewater is $\approx 286 + 150 + 60 + 139 = 635$ mg/l

BOD computation - For most readily degradable wastewaters, the ultimate BOD, BOD_U, is ≈ 0.92 COD. For the wastewater in question, phorone (50 mg) and the sulphide (30 mg) are not easily degradable, therefore only 220 mg (≈ 73 %) of the wastewater is easily degradable.

$$BOD_U \approx (0.73)(0.92)(COD) = 0.73 \times 0.92 \times 635 = 426.5 \text{ mg/l}$$

$$BOD_u = L_o(1 - e^{-20k})$$

Using the typical value of k, 0.23 day-1

$$\begin{aligned} BOD_u &= L_o(1 - e^{-20(0.23)}) \\ &= L_o\,(1 - 0.01) \\ &= 0.99L_o \\ 426.5 &= 0.99L_o \end{aligned}$$

$$L_o = 430.8$$

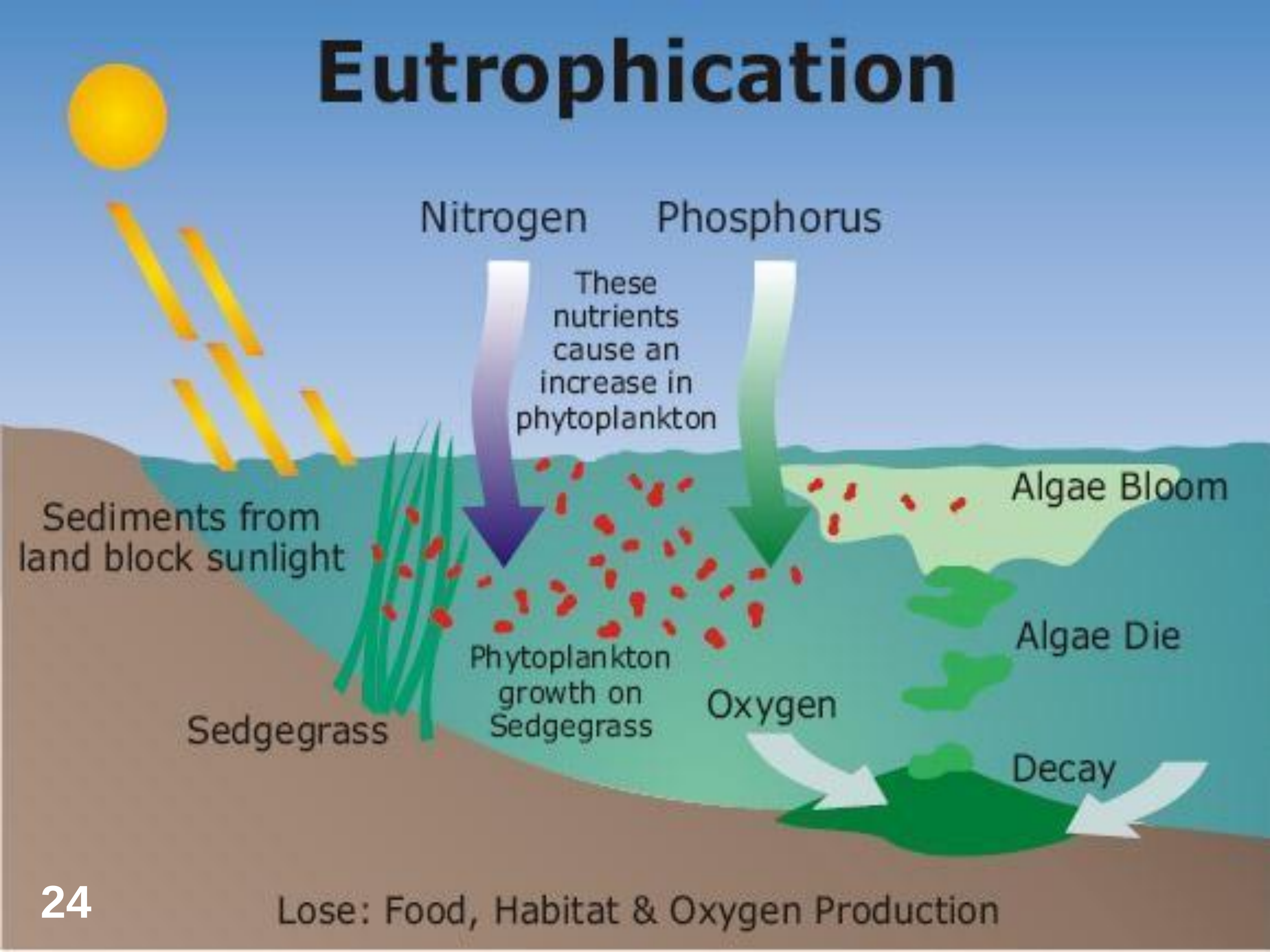
$$\begin{aligned} BOD_5 &= L_o(1 - e^{-5(0.23)}) \\ &= 430.8\,(1 - 0.32) \\ &= 294.4 \text{ mg/l} \end{aligned}$$

For the wastewater in question:

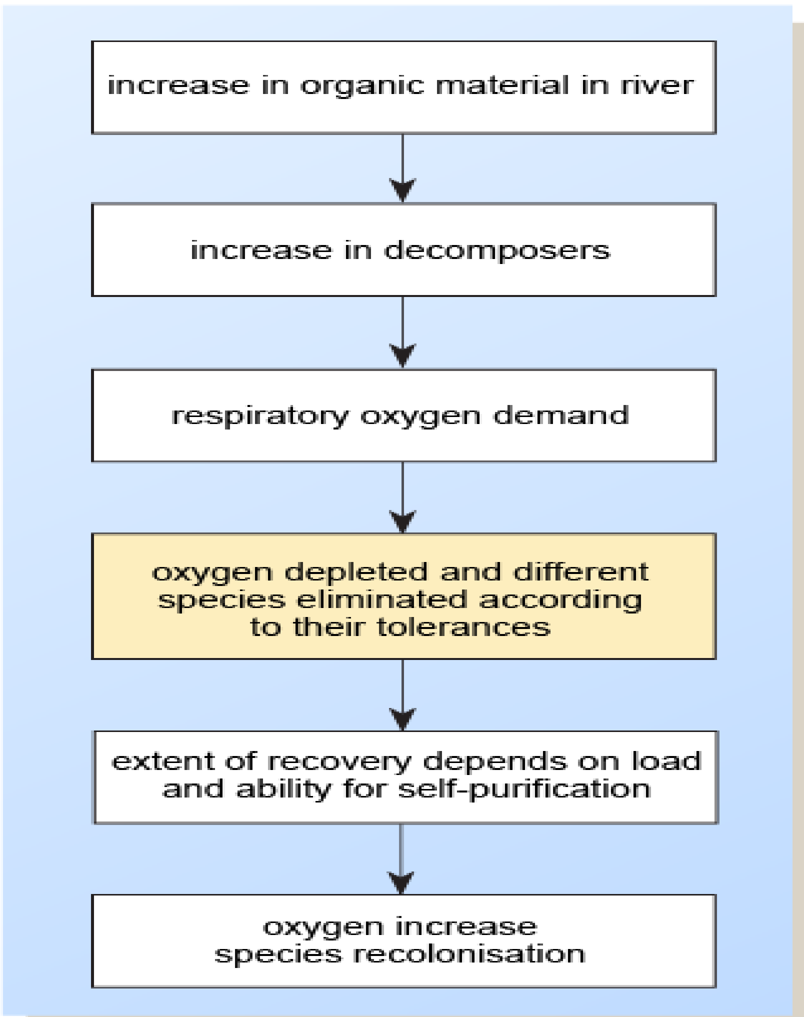
TOC = 168.5 mg/l, COD = 635.0 mg/l, BOD = 294.4 mg/l

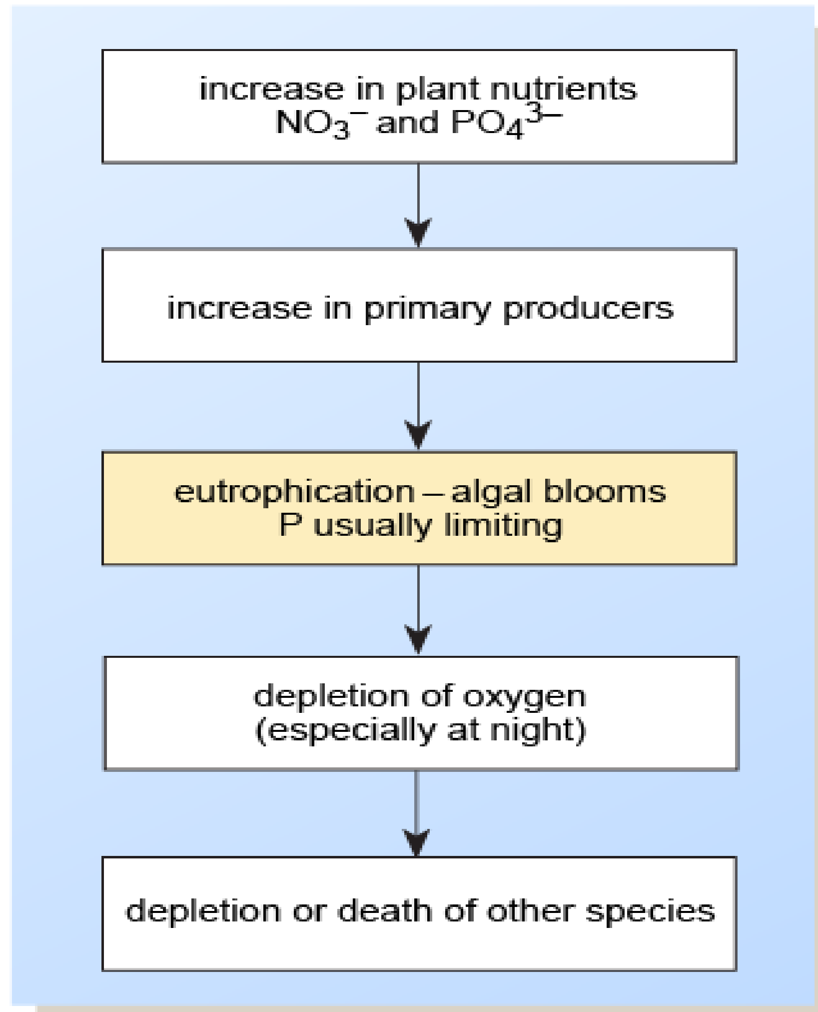
Chemical Characteristics – contd.

- Oil & Grease
- Total Nitrogen & Total Phosphorus
  - Responsible for growth of aquatic life ->algal bloom -> eutrophication. (Plants require this nutrients in very small concentrations, when present in large amounts, they fertilize the water for algae to grow. This excessive growth of algae is known as algae bloom. This covers the water body and hinder sunlight from getting to organisms at the bottom, thus hindering their growth. On dying they decay and deplete the oxygen which eventually leads to fish kill, and eutrophication of the water body.
- Priority Pollutants (Toxic pollutants)
  - These are compounds selected on the basis of their known or suspected carinogenicity, mutagenicity and teratogenicity, they include: Nitrates, cyanides and heavy metals

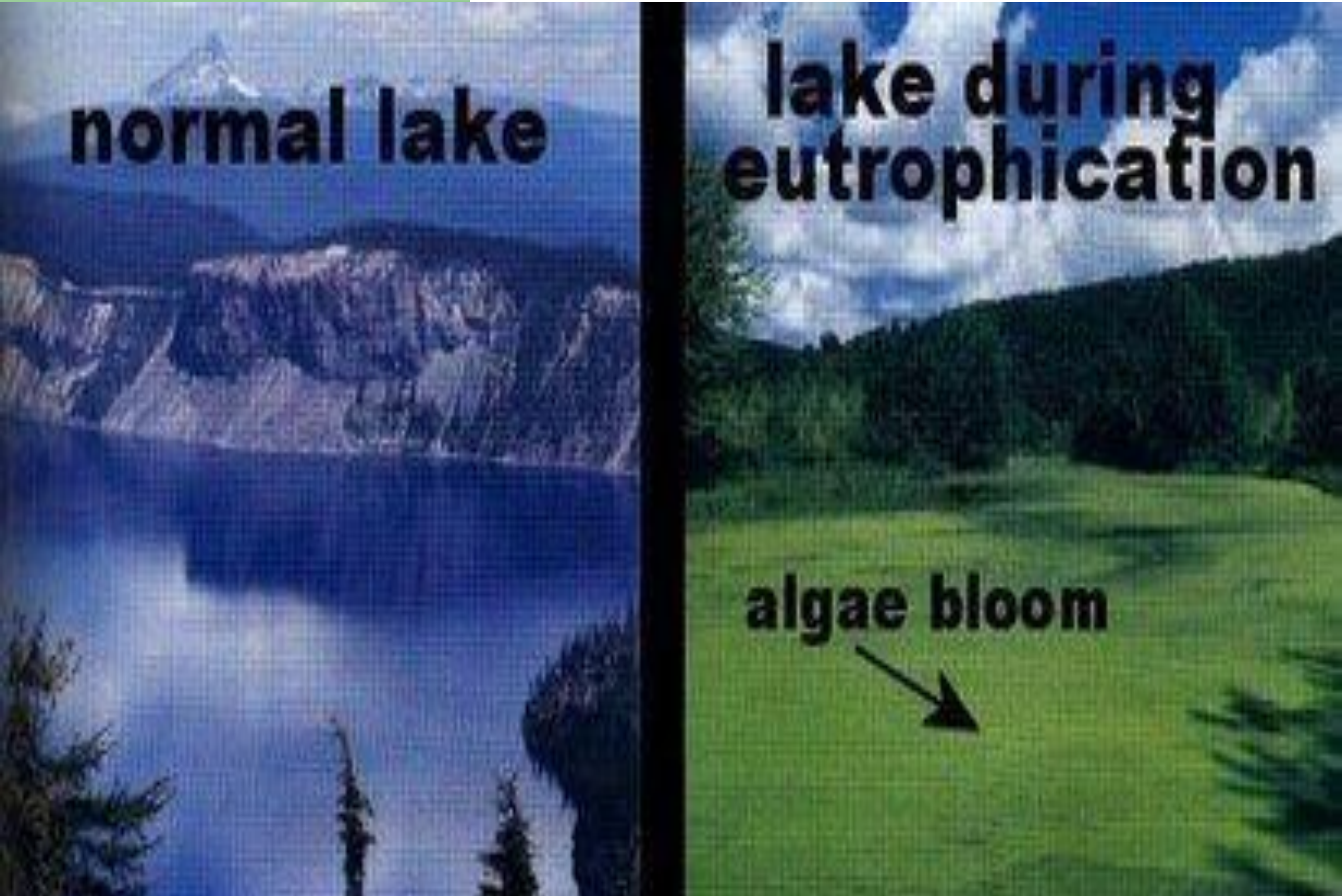

Eutrophication
Nitrogen
Phosphorus
These nutrients cause an increase in phytoplankton
Algae Bloom
Sediments from land block sunlight
Algae Die
Phytoplankton growth on Sedgegrass
Oxygen
Sedgegrass
Decay
Lose: Food, Habitat & Oxygen Production

Flowchart for
Organic Pollution / Eutrophication





Eutrophication



Why are lakes more susceptible to eutrophication?

Toxic Pollutants

- **Heavy metals**

  atomic wt. >23 and sp,gr >5

  In terms of environmental impact, most important ones are:
  - **Mercury,**
  - **Lead,**
  - **Cadmium, and**
  - **Arsenic**

- Nitrates

  in babies, nitrates get converted to nitrite by the bacteria in the digestive system. These nitrites combine with haemoglobin in the bloodstream - (met- haemoglobin-aemia) and inhibit oxygen distribution around the body – **BLUE BABY SNYDROME**

- Cyanide also attaches to the Hb, impairing oxygen transport

Toxic Pollutants

Toxicity of materials depends on pH:

- **Nickel cyanide complex is 500 times more toxic to fish at pH 7 than at pH 8 bcos the complex dissociates into cyanide and nickel ions. Cyanide forms hydrogen cyanide HCN**
- **Ammonia is 10 times more toxic at pH 8 than at pH 7**

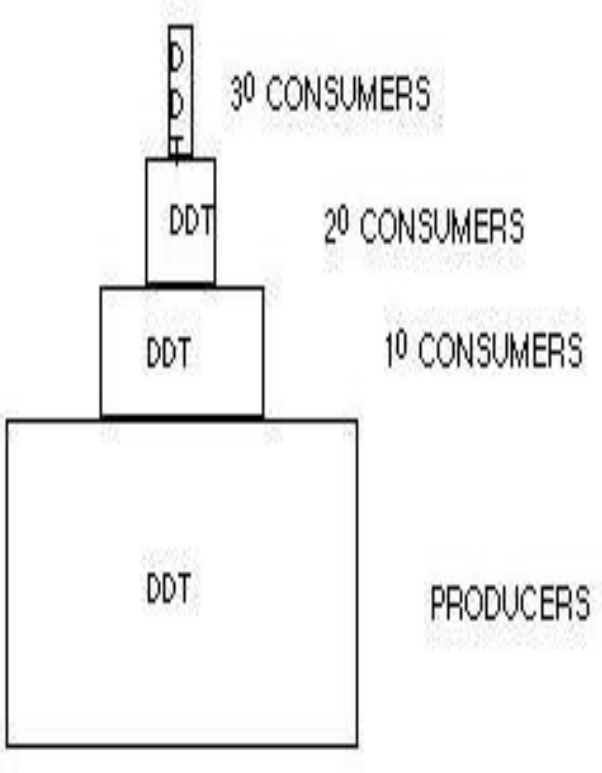
The two significant groups of toxic pollutants are:

- **Heavy metals, and**
- **Synthetic Organic substances – do not easily biodegrade naturally, e.g pesticides (DDT), therefore bioaccumulates – remains in the body, not broken down by digestion, and biomagnifies, up the food chain. Also called Persistent (stubborn) Organic Pollutants**

concentration increases 10 fold as biomass is transferred from one level of the food chain to the next

The concentration of DDT tends to increase in food chain from one level to the next. Those at the top of the food chain receive the largest doses.

Biomagnification of DDT is due to:
1. high solubility in tissue fats; low in water;
2. very low rate of metabolism; low loss from body;
3. biomass transfer from one level of the food chain to next.



Toxicity of some chemicals in small mammals

Toxicity is measured by LD_{50}

- **i.e. Lethal dose large enough to kill 50% of the sample of animals under test**

LD_{50} (mg per kg body weight)	Examples	Classification
1–10	arsenic	highly toxic
10–100	cadmium copper lead mercury	moderately toxic
100–1000	aluminium molybdenum zinc	slightly toxic
>1000	sodium iodine calcium potassium	relatively harmless

Observed effect of pH of water on fish

pH range	Effect
6.5–9.0	No effect
6.0–6.4	Unlikely to be harmful except when carbon dioxide levels are very high (1000 mg l^{-1})
5.0–5.9	Not especially harmful except when carbon dioxide levels are high (20 mg l^{-1}) or ferric ions are present
4.5–4.9	Harmful to the eggs of salmon and trout species (salmonids) and to adult fish when levels of calcium, sodium and chloride are low
4.0–4.4	Harmful to adult fish of many types that have not been progressively acclimated to low pH
3.5–3.9	Lethal to salmonids, although acclimated roach can survive for longer
3.0–3.4	Most fish are killed within hours at these levels

Biological Characteristics

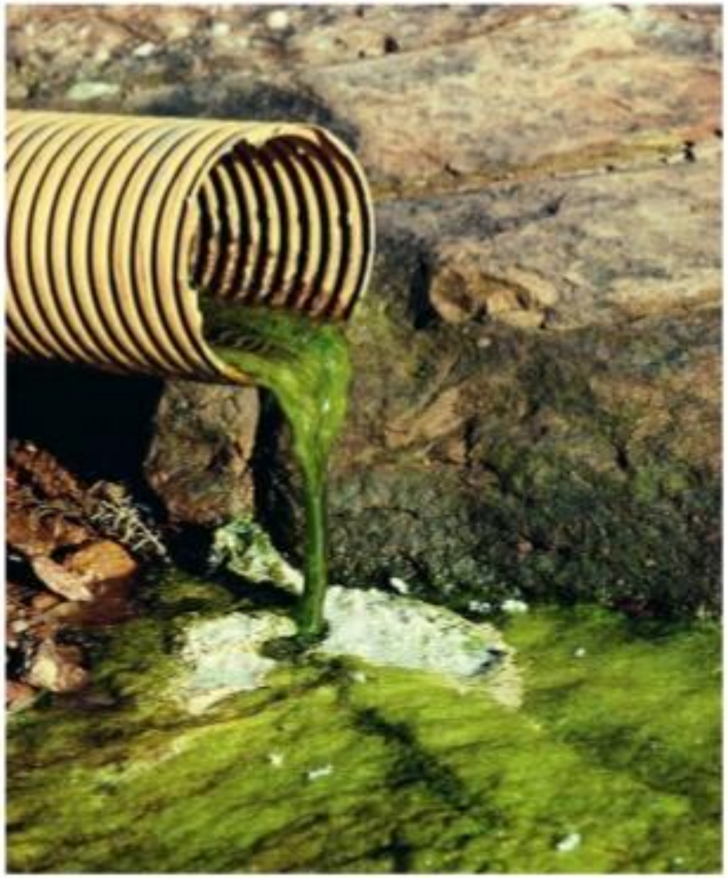
- Bacteria
- Viruses
- Fungi
- Protozoa
- Algae

Biological pollutants can spread disease through water and disrupt ecology. The total coliform test is used for biological characteristics estimate

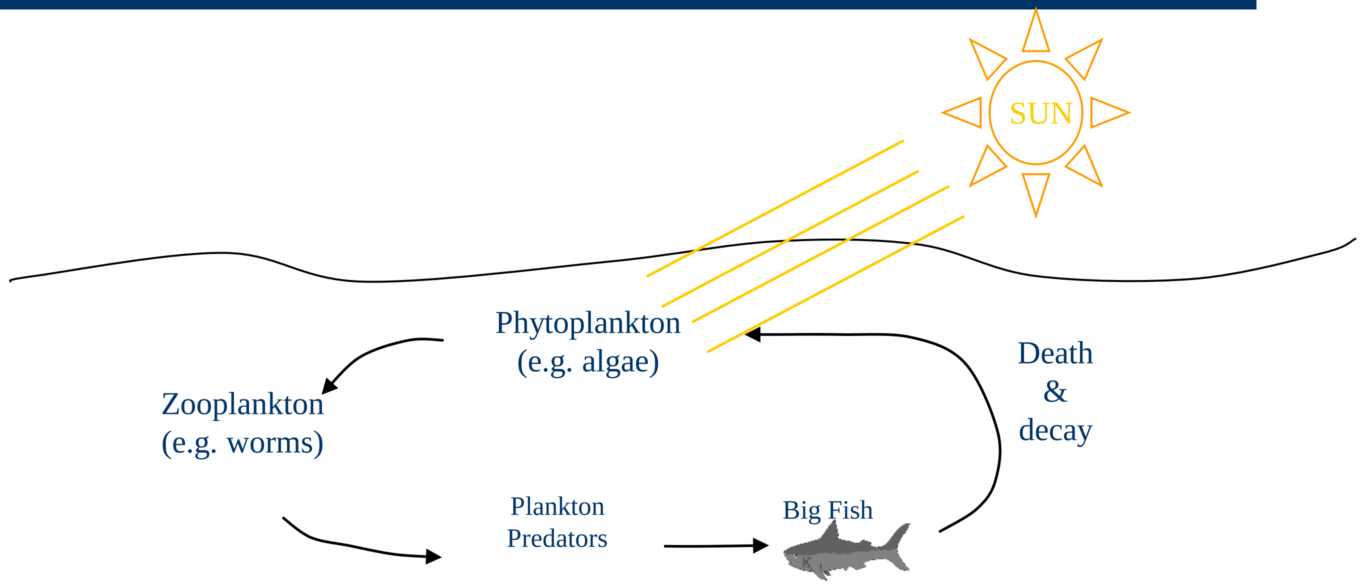
Typical Industrial Wastewater Effluent Limitations

Parameter	Concentration (mg/L)
COD	300-2,000
BOD	100-300
Oil & Grease/TPH	15-55
TSS	15-45
pH	6.0-9.0
Temperature	< 40 °C
Color	2 color units
NH_3/NO_3^-	1.0-10
Phosphates	0.2
Heavy Metals	0.1-5.0
Surfactants	0.5-1.0 (total)
Sulfides	0.01-0.1

Water Pollution Pictures

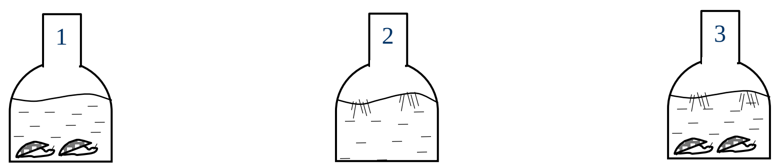


EFFECT OF WASTE WATER CHARACTERISTICS ON THE WATER ECOSYSTEM



A simplified water food chain

A Simple food chain experiment



	1	2	3
	Snail No Algae	Algae No Snail	Algae Snail
Result	Death	Death	Both survive

Biogeochemical cycles of life

Life in general is made up of chemical elements, which exists in the right amounts (quantity), the right concentrations (quality), and the right ratios to one another.

The big six macronutrients (they form the fundamental building blocks of life) **are:**

Carbon - along with H and O form carbohydrates

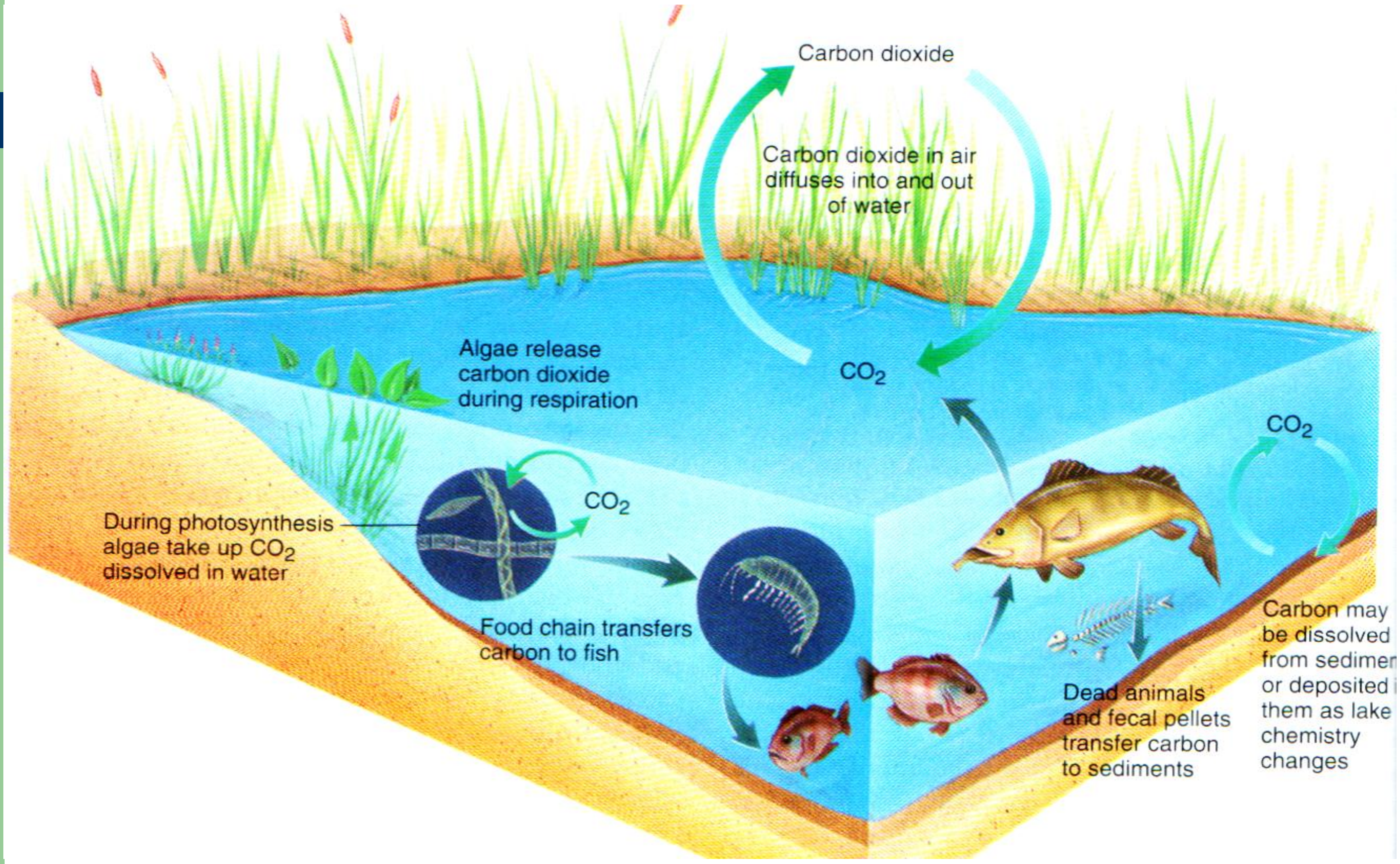
Nitrogen - along with C, H and O form proteins

Phosphorus - energy element, used for the transfer and use of energy within cells

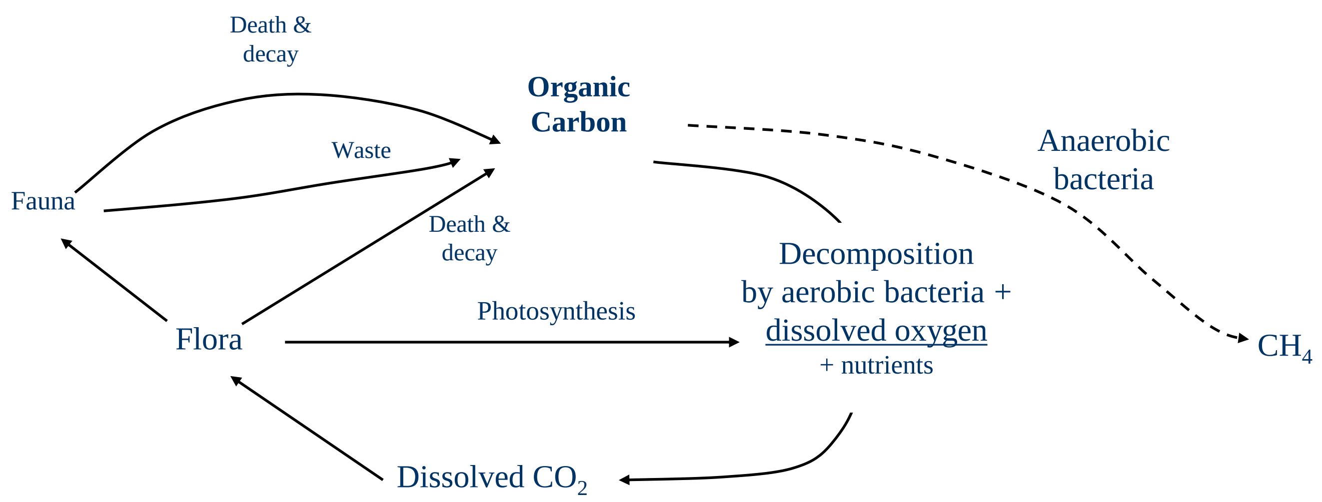
Others are Calcium - structure element

Sodium, Potassium - for nerve signal transmission

Carbon cycle in the water body

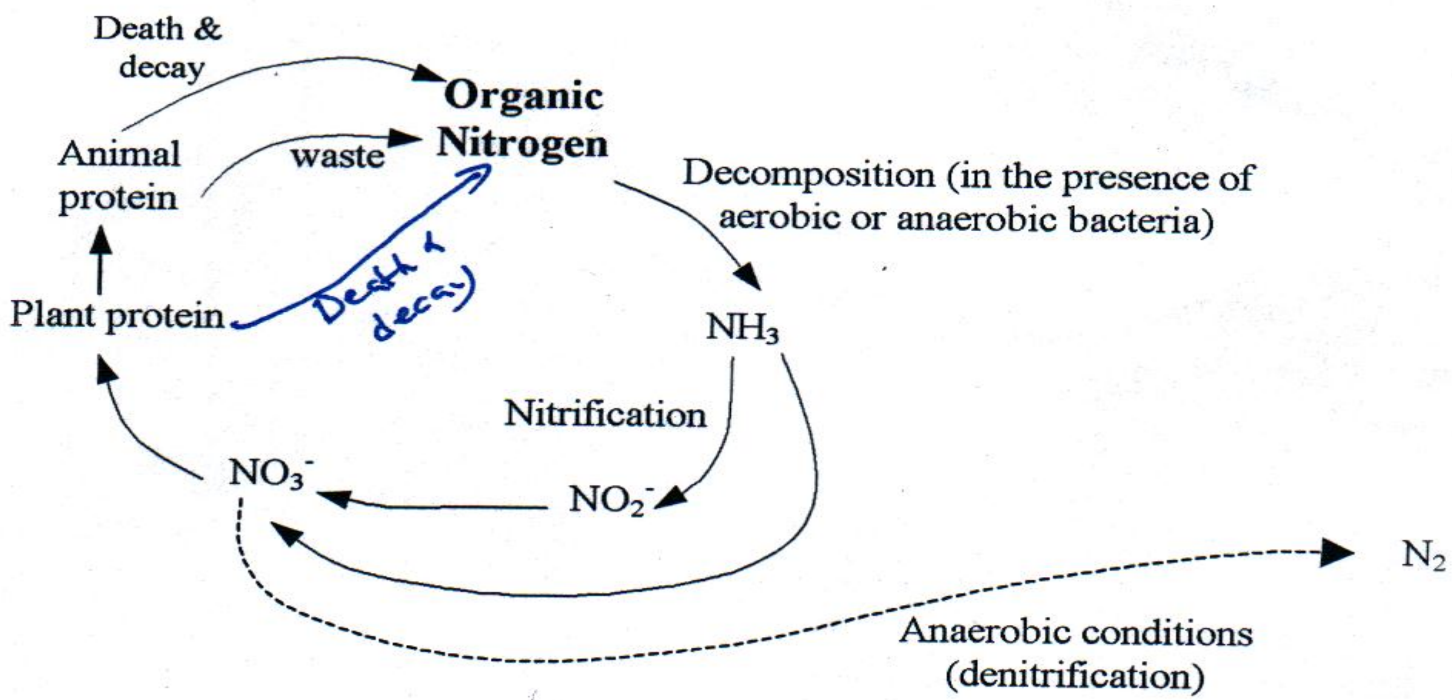


The carbon cycle and the effect of wastewater



What's the effect of adding wastewater with high suspended solids and high BOD?

The Nitrogen cycle and the effect of wastewater



Summarized effect of different pollutants based on their classifications

Pollutant	General effect	Effect on biota	Effect on water supplies	Sources: natural	Sources: resul of human activity
Organic (biodegradable wastes)	Increased oxygen demand; food provided for organisms lower down in food chain	Tolerated in moderate quantities if release not too quick, serious if dissolved oxygen (DO) drops too quickly	Increased need of treatment	Run-off and seepage through soil	Domestic sewage, food processing, animal wastes
Plant nutrients	Excessive plant growth	Demand on DO	Increased need of treatment	Natural degradative processes	Animal wastes, fertilisers, detergents, industrial waste
Toxic chemicals (e.g. heavy metals, pesticides, phenols, PCBs)	Toxic to humans, animals and plants	Could be lethal	Increased need of treatment or control	Rare	Detergents, pesticides, tanneries, pharmaceutica, wool scouring refineries

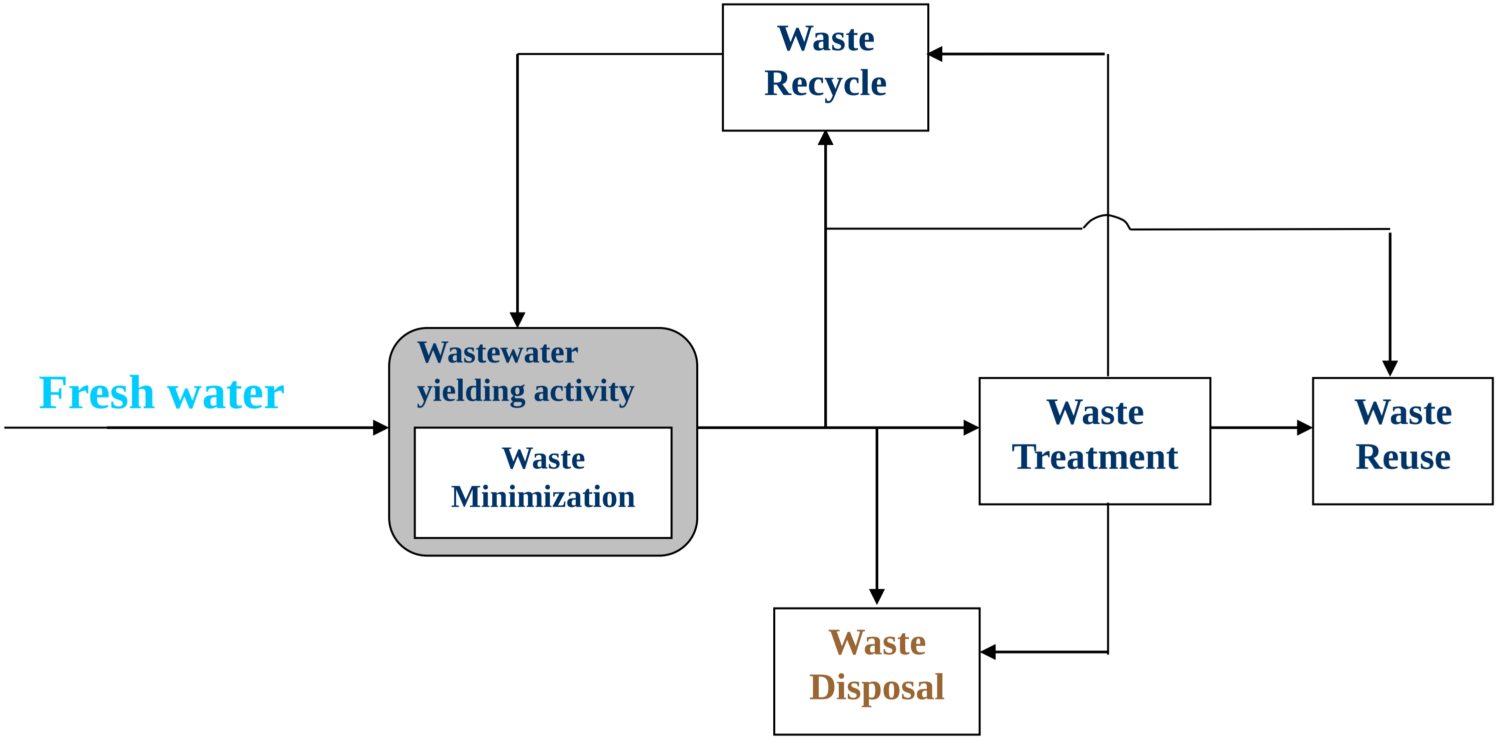
Endocrine disruptors	Alteration of ecology	May adversely affect health and reproduction of humans and animals	Can be present in water sold in plastic bottles	*Fusarium* species of fungus	Chemical manufacture, intensive farming
Acids/alkalis	Lowering/ raising of pH; acids can dissolve heavy metals	Only narrow range of pH tolerable for most plants and animals; heavy metals toxic	Corrosion	Naturally acid or alkaline rock	Battery, steel, chemical and textile manufacturing, coal mining
Suspended solids	Reduction in light penetration (increased turbidity), blanketing, introduction of colour	Photosynthesis reduced; blanketing of benthic plants and animals; obstruction of gills of fish	Obstruction of filters; increased need of treatment	Soil erosion, storms, floods	Pulp mills, quarrying, any building or development work involving ground disturbance

Immiscible liquids	Formation of a layer at the water surface that could prevent O_2/ CO_2 interchange	Reduced DO; insect breeding affected	Interference with treatment processes	Unlikely	Oil-related activity
Heat	Decrease in DO; increase in metabolic rate of aquatic organisms	Possible reduced breeding or growth of aquatic organisms	None	Unlikely	Power plants, steel mills
Taste-, odour- and colour-forming compounds	Taste, malodour, colour	Tainting of fish	Increased need of treatment	Peat	Chemical manufacture or processing
Microorganisms	Pathogenic to humans	None	Increased need of treatment	Animal excrement	Contamination from human wastes

Management Options

- **Disposal –** Simple discharge to nearby water bodies
- **Treatment -** an end-of-pipe approach to the waste problem. Medicine after death because once pollutants have been formed, one is simply just moving them from one place to another – there is still an environmental problem
- **Recycle/ Reuse –** cradle-2-cradle/circular economy
- **Minimization -** Wastewater minimization is based on the precautionary principle: Prevention is better than cure. Seeks to prevent, eliminate or reduce waste at its source
- **Prevention** – No wastewater generation (e.g by shifting to a waterless process where possible)

Pathways in Wastewater Management



WASTEWATER TREATMENT

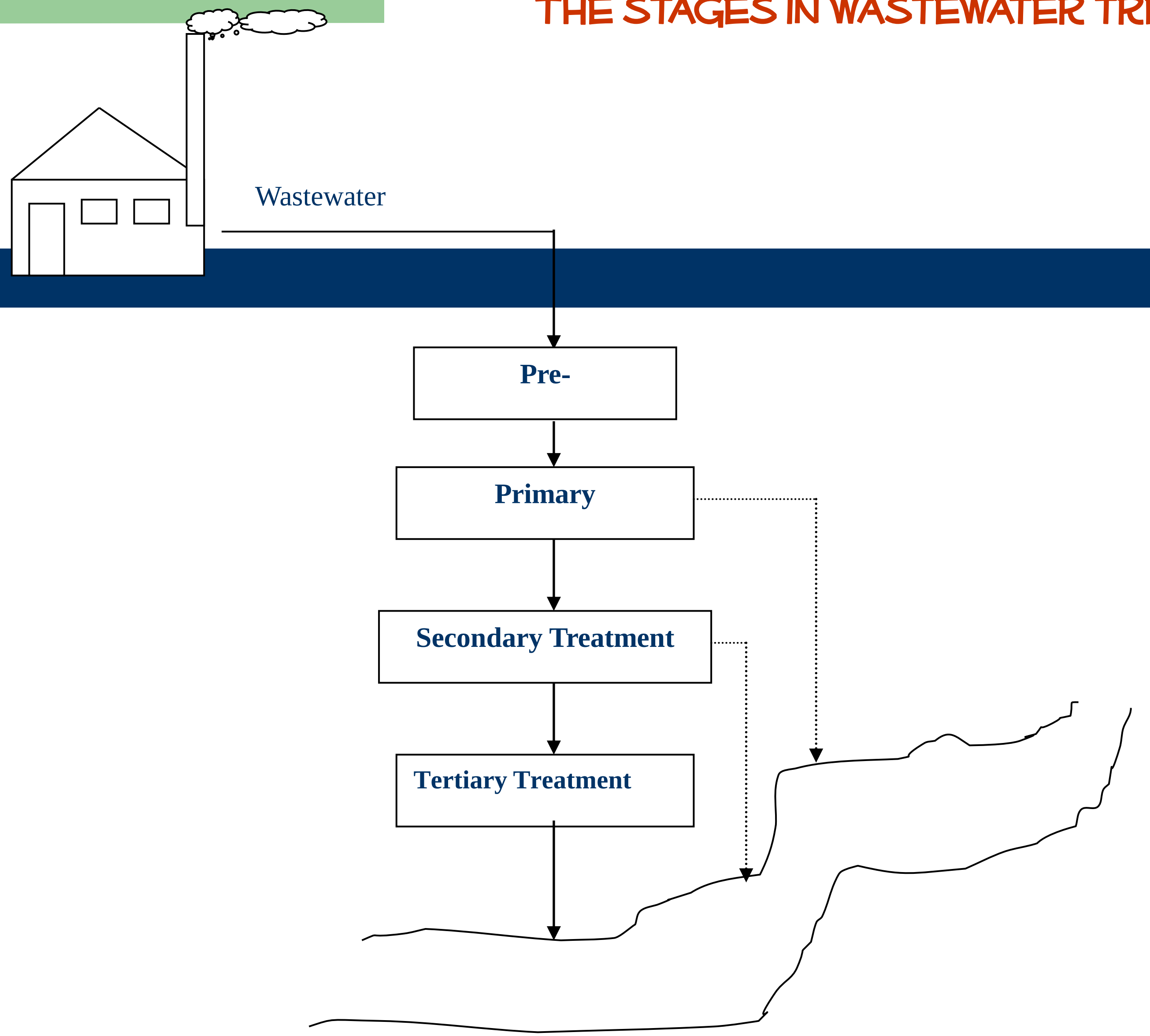
For effective treatment, the characteristics of the wastewater must be juxtaposed with the target, which is usually the acceptable limit set by the environmental body, and also the intended end use of the treated wastewater . This will guide in selection and design of the appropriate treatment processes.

TYPES OF TREATMENT

- **physical**
- **chemical**
- **biological**

In treating wastewater a combination of these methods are used as the treatment process progresses from the pre-treatment stage to the tertiary treatment stage. Each treatment stage is aimed at removing a more specific class of pollutants.

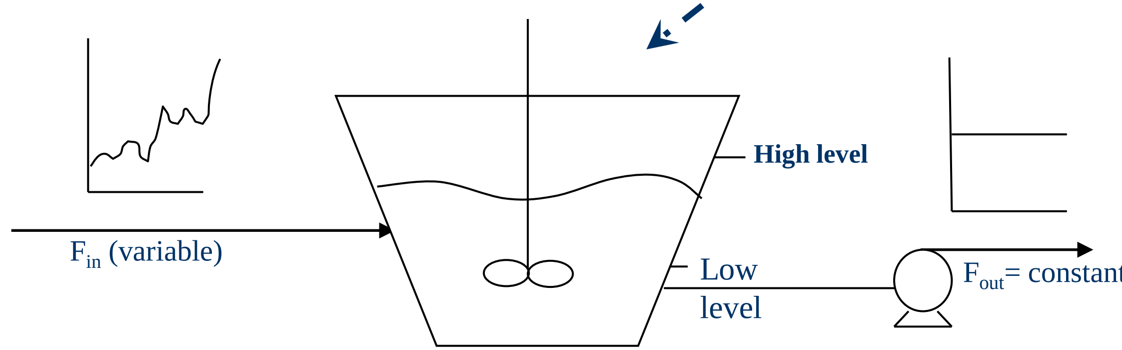
THE STAGES IN WASTEWATER TREATMENT



PRETREATMENT STAGE

remove large objects usually suspended in the wastewater, and floating materials that might cause damages to the equipment in the treatment plant

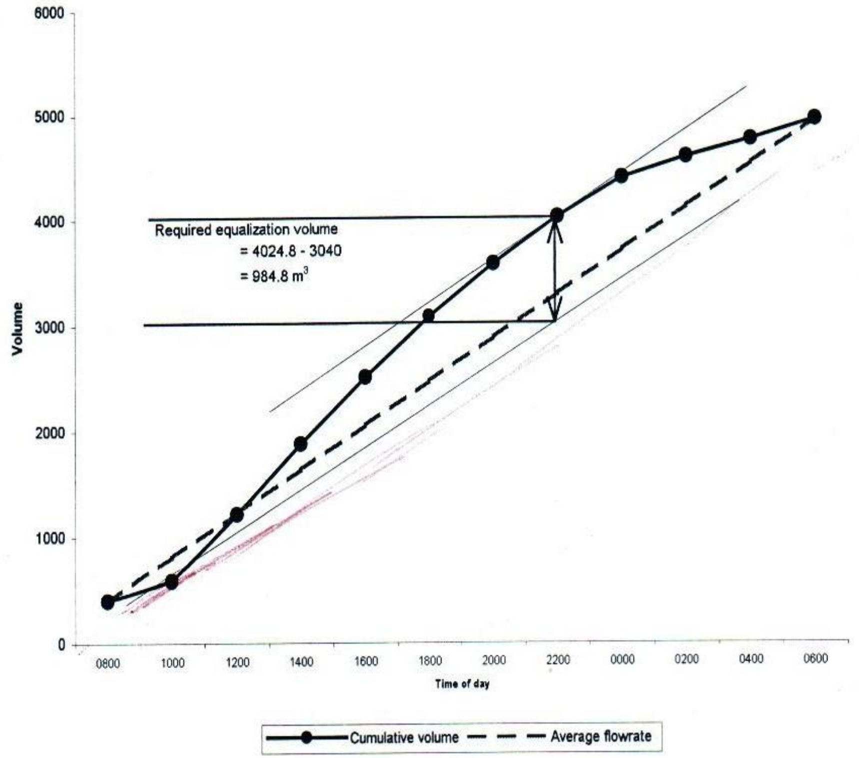
→ Screening → Grit Removal → Comminution → Equalization →



Design of Equalization tank

Graphical or analytical approach

- **Plot cumulative volume into tank versus time**
- **Avg. Daily flow is line from origin of cumulative plot to end**
- **Draw tangent, at low point – denotes empty tank, and tangent at high point denotes full tank.**
- **The distance btw these two tangent lines gives the required volume of the tank**



Primary treatment Stage

The main objective of primary treatment is the **removal of solids** dissolved, suspended or colloids), **oil and grease** from the wastewater.

- **Suspended solids** (density > wasterwater) are removed by **sedimentation**
- Dissolved solids (organics or minerals) are **precipitated out** and removed by sedimentation or **filtration**.
- **Colloids** by **coagulation** followed by **flocculation**
- **Oil and grease,** solids (density < wastewater), are removed by **flotation**

For the operations or processes mentioned above to be effective an optimum pH is required. It is therefore necessary to **adjust the pH or neutralise** the wastewater before proceeding with treatment

Sedimentation tanks

can be

- **Circular with radial flow**
- **Rectangular with horizon flow**

Removes 70% Suspended solids , bcos some of these solids are biodegradable, Removes 30% BOD

$$\text{retention time} = \frac{\text{Volume of sedimentation tank}}{\text{flow rate through sedimentation tank}}$$

Typical retention time = 2 - 6 hrs Typical depths = 2 – 4 metres

Precipitation

In industrial wastewaters undesirable heavy metals are present in soluble form, these are best removed by converting to insoluble forms by chemical reaction - **(see Table on slide 47)**

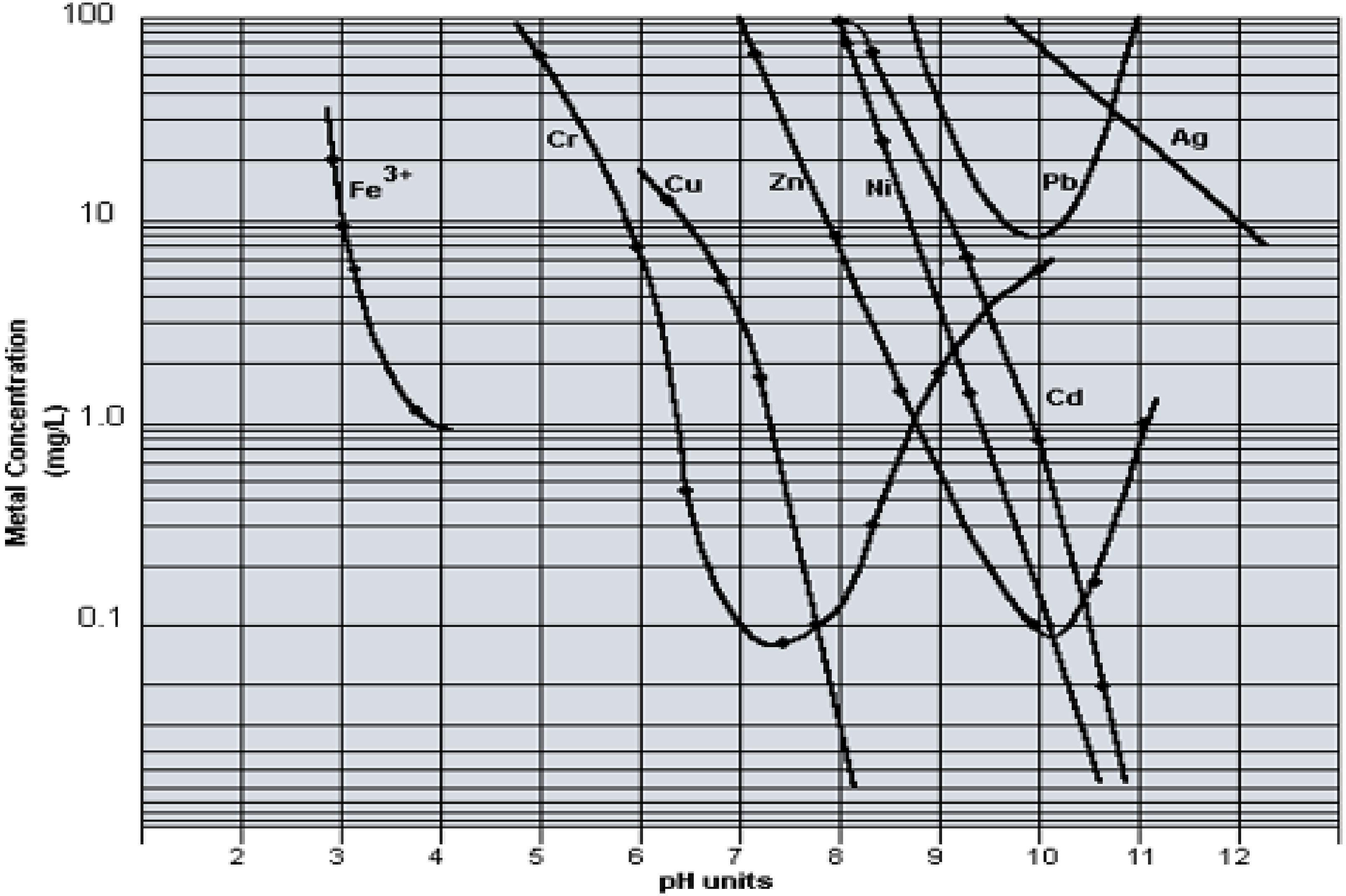
Metals precipitate at different pH,

e.g. Chromium-pH 7.5, Zinc – 10.2 **(see plot on slide 55)**

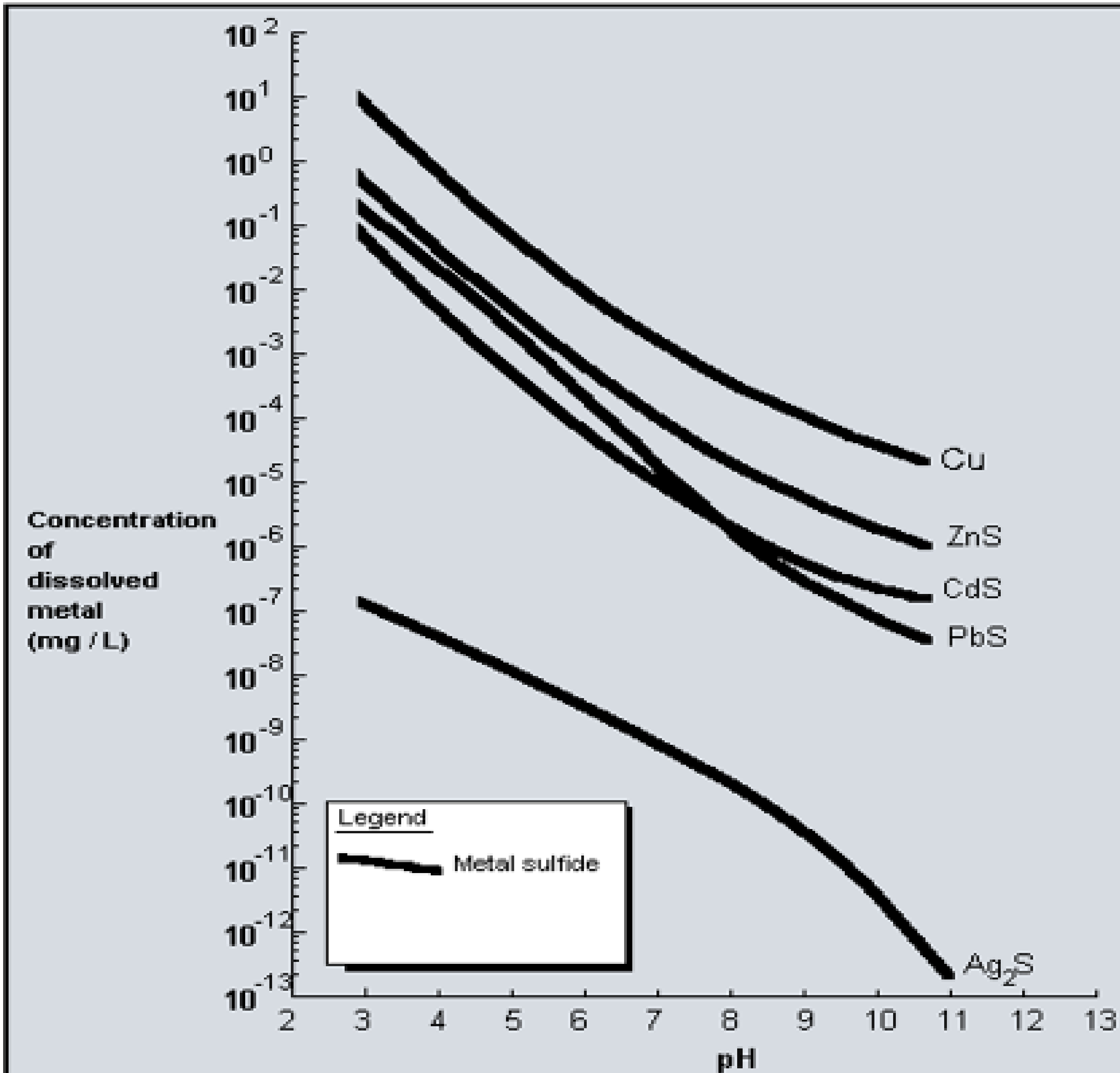
Chromium (VI) ion is reduced to Chromium (III) ion using Sulphur dioxide

Solubility of metals versus pH

The most common methood to remove soluble metal ions from solition isto precipitate the ion as a metal hydroxide. Below is the metal hydroxide solubility curve.
pH can be raised with a common alkaline material eg lime or sodium hydroxide



Sulphide solubility chart



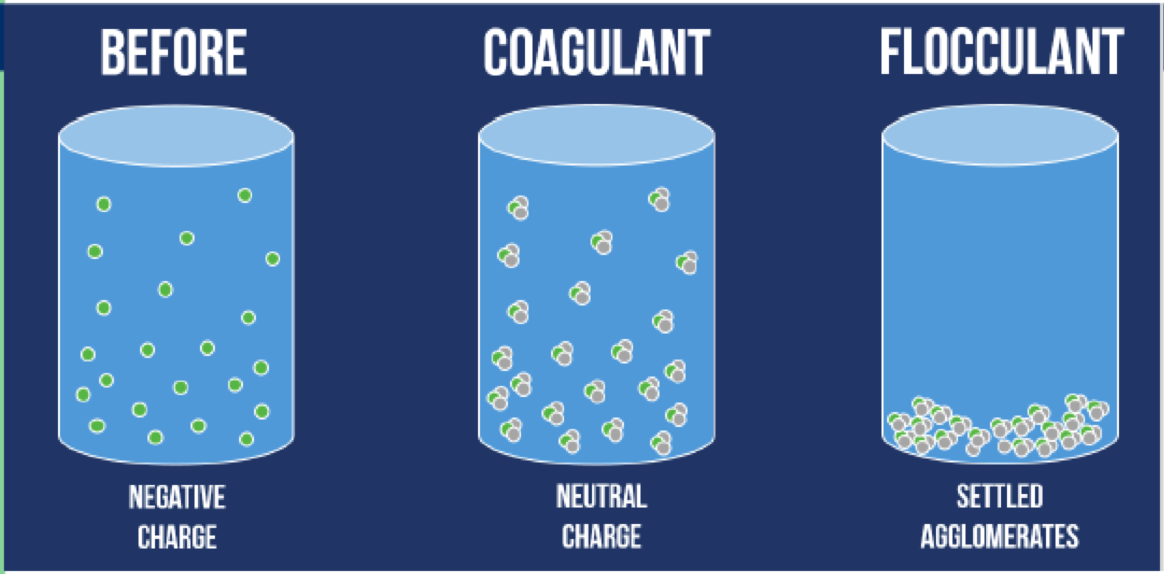
Precipitation of heavy metals

Metal	Sources	Optimum pH and means of precipitation	Inhibitors	Other removal methods
Arsenic	Glassware, ceramic, tannery, dye, pesticide, metallurgical, petroleum refining, chemicals industries	Precipitated as the sulphide by the addition of Sodium or Hydrogen Sulphide at pH of 6 – 7.		
Barium	Paint and Pigment, glass, ceramic, dye, vulcanizing of rubber and metallurgical industries	Precipitated as barium sulphate (which is extremely soluble) at an optimum pH of 10		Ion-exchange Electrodialysis
Cadmium	Metallurgical alloying, electroplating, ceramics, photography, pigment, textile printing and chemical industries	Precipitated as the hydroxide at pH range of 9.5 – 12.5	Cyanide	
Chromium	Ink manufacturers, dyes & paint pigments, chrome tanning, metal cleaning, plating and electroplating operations.	Precipitated as chromic hydroxide $Cr(OH)_3$ by the addition of lime at pH 8.5 – 9.5 [Chromium in the form of chromate (CrO_4^{2-}) or dichromate ($Cr_2O_7^{2-}$) must be reduced to the trivalent form Cr^{3+} before it can be removed by precipitation].		
Copper	Metal-process pickling and plating baths, chemical processes using copper salts or copper catalyst.	Precipitated as hydroxide at an alkaline pH (9.5 – 12.5)by the addition of sodium hydroxide	Calcium Sulphate, cyanide, ammonia	Ion-exchange Electrodialysis Evaporation
Cyanide*	Electroplating industry	Completely oxidized to carbon dioxide CO_2 and nitrogen N_2 by chlorination (direct addition of Sodium hypochlorite) at an alkaline pH		Electrolytic decomposition
Fluorides*	Glass, electroplating, steel, aluminum, pesticide and the fertilizer industries.	Precipitated as calcium fluoride by the addition of lime.		
Iron	Wastewaters from Chemical industries, dye, metal processing, textile mills, mining operations, ore milling and petroleum refining industries.	At neutral pH of 7 and in the presence of oxygen (achieved by aeration), soluble ferrous iron gets oxidized to ferric iron which readily hydrolyzes to form the insoluble ferric hydroxide precipitate		
Lead	Battery manufacturers	Precipitated as lead carbonate $PbCO_3$ by the addition of soda ash, $NaCO_3$ at pH 9.0 –9.5, or as lead hydroxide $Pb(OH)_2$ by the addition of lime at pH 11.5, or as lead sulphide by addition of sodium sulphide at pH 7.5 – 8.5		
Mercury	Chlor-alkali industry, electrical & electronics, explosive manufacture, photographic, pesticide, preservative, chemical & petrochemical industry (where it is used as a catalyst).	Precipitated as the highly insoluble mercury sulphide by the addition of sodium sulphide at a pH of 8.5		Coagulation with alum, iron salts and lime.
Nickel	Metal processing, motor vechicles, aircrafts, printing and some chemical industries	Precipitated as an hydroxide upon addition of lime at pH 10 – 11	Cyanide	
Selenium	Paper, pigment & dye and in fly-ash	Precipitates as sulphide at a pH of 6.6		
Silver	Porcelain, photographic, electroplating and ink manufacture.	Precipitates as a chloride.	Cyanide	
Zinc	Steel works, fiber manufacture, plating & metal processing industries.	Precipitates as an hydroxide by the addition of lime or caustic.		

* Even though usually classified under heavy metals (which are positively charged), they are anions (negatively charged)

Colloid removal

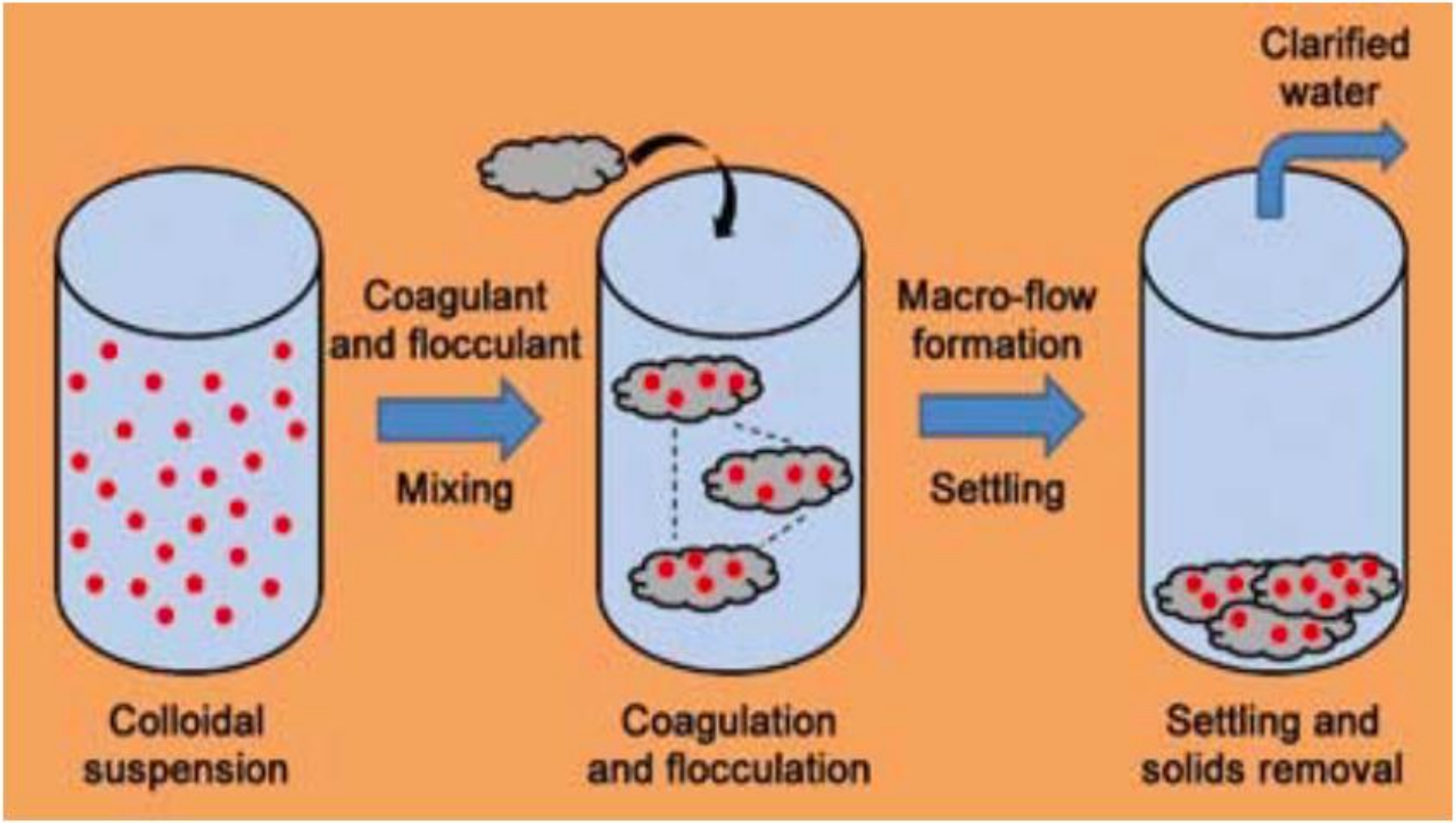
-by Coagulation and Flocculation



Coagulation is the addition of **coagulant** to neutralize charged particle. **Flocculation** is a mixing technique that promotes agglomeration and assists **in the** settling of particles.

Alum (aluminum sulphate) is a good example of a coagulant

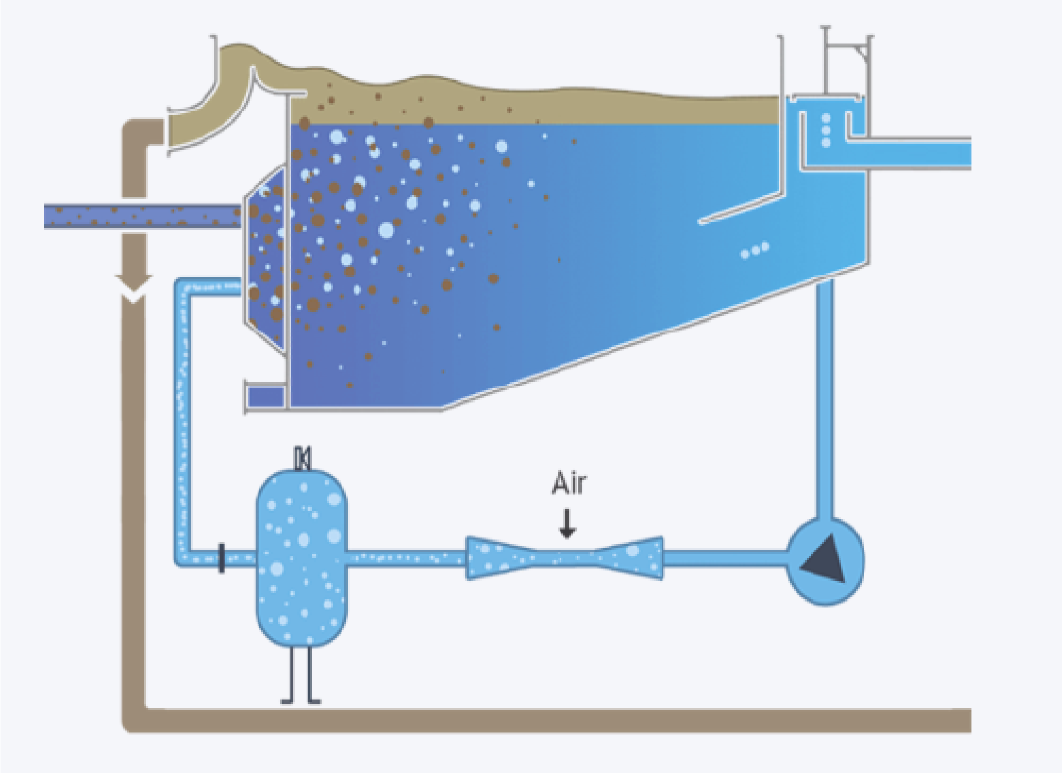
Coagulation and Flocculation



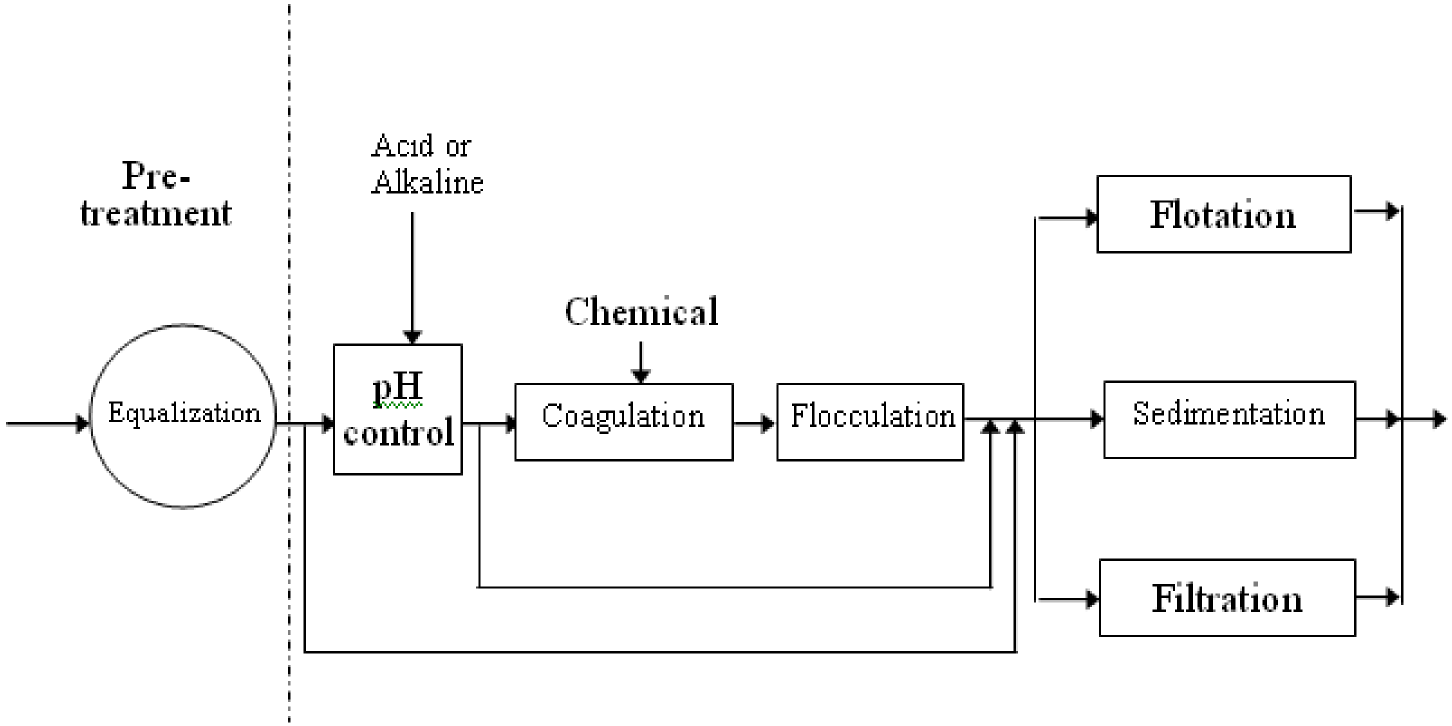
Removal of Fats, oil & Grease (FOG)

- By Dissolved Air Flotation

Wastewater is pumped into a flotation tank or basin and hit with dissolved air causing microscopic bubbles to rise to the surface of the water, as they do so FOG emulsified in the wastewater adhere to the bubbles and thereby rise to the surface where they may then be removed by a skimming device and drained toward a sludge discharge tank.



Alternative operations and processes for the primary treatment stage of industrial wastewaters



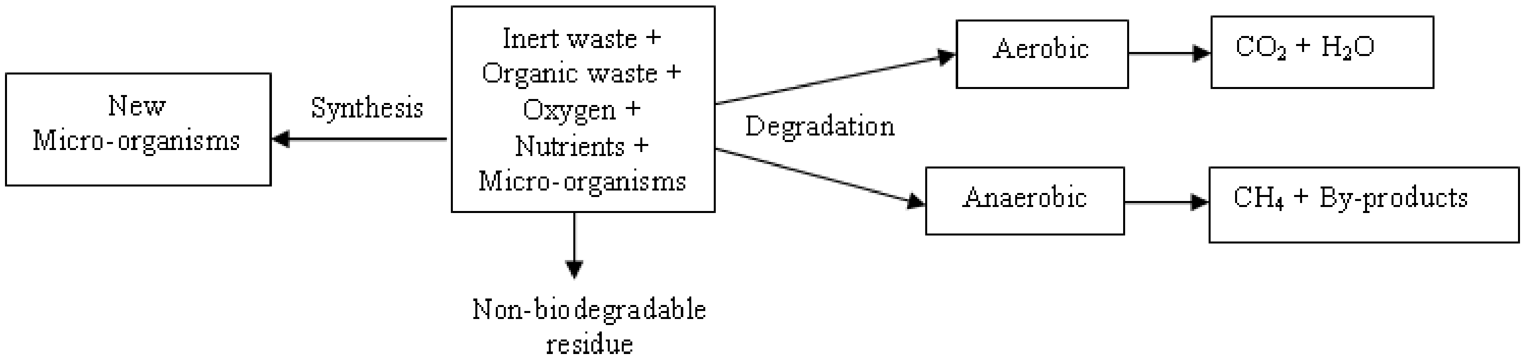
Secondary Treatment Methods

In biological treatment processes, microorganisms such as bacteria, fungi, protozoa and algae are used to break down the soluble organics (measured as BOD) in the wastewater. These organisms, in carrying out their metabolic activities, feed on the soluble organics and in doing so decomposes them to carbon-dioxide, water, energy, non-biodegradable residue and at the same time the organisms grow and produce more biological cell tissues

- Factors that make for effective secondary treatment include:
  - Elimination of shock loadings via the use of equalisation tanks
  - Maintaining an optimum temperature. 15 – 30oC,
  - Maintaining an optimum pH 6.5 – 8.5.
  - Absence of high concentrations of toxic metals:
  - Absence of excessive oil & grease (coats the surface of the microorganisms thereby hindering oxygen transfer)

Types of Biological Treatment Processes

- **Aerobic processes**: microorganisms use oxygen for metabolism,
- **Anaerobic processes**: microorganisms do not require oxygen for metabolism.

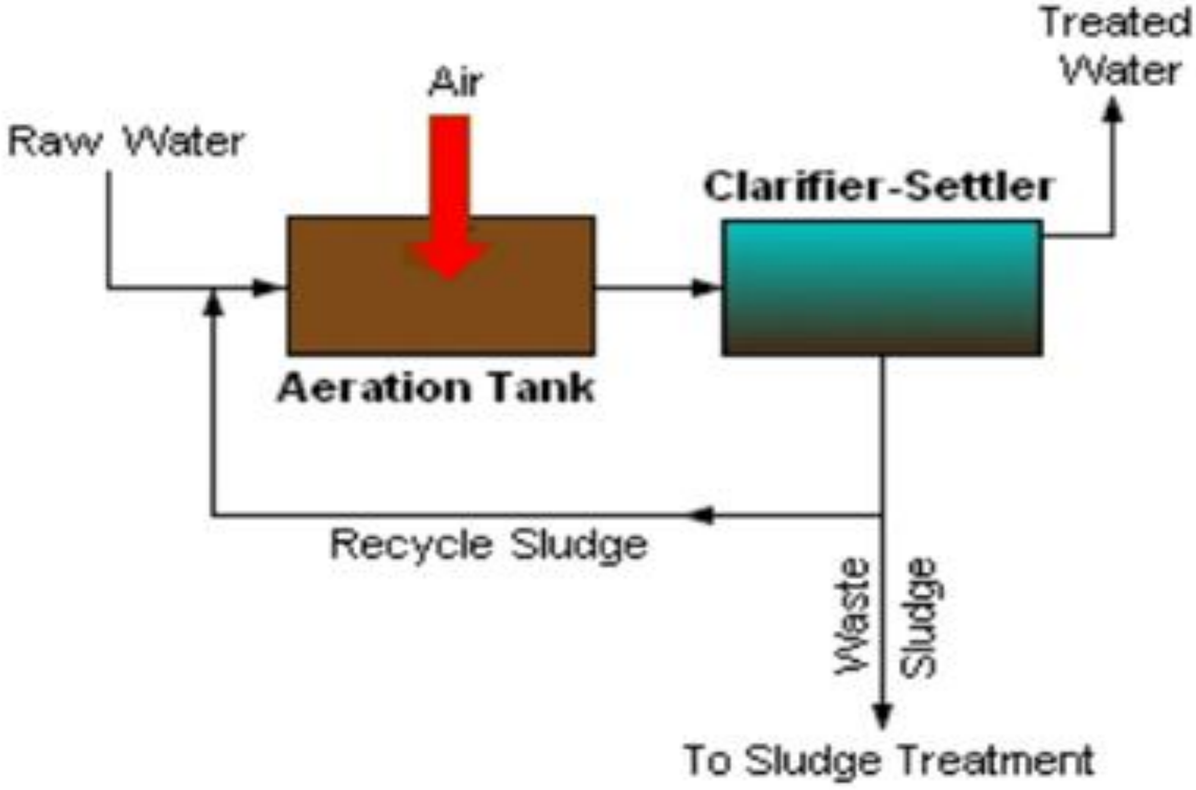


Types of Aerobic Treatment Processes

- **Suspended-growth systems**: in which case the micro- organisms are in suspension in the waste water, and

- **Attached-growth or fixed-film systems,** where the microorganisms are attached to some inert medium, such as rocks, specially designed plastics or ceramics, and the wastewater is made to flow over the medium.

SUSPENDED GROWTH SYSTEM :
ACTIVATED SLUDGE PROCESSES



Organic loading rate

Principal parameter is the organic loading rate -

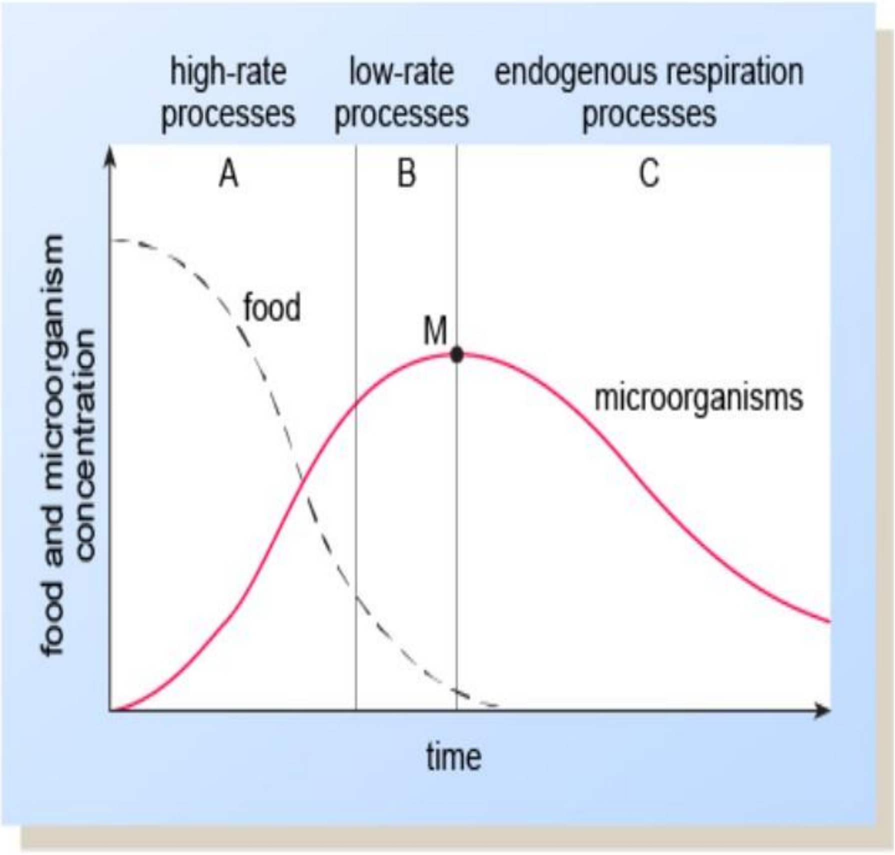
ratio of food to microorganisms (F/M)

$$F/_M = \frac{\text{mass of BOD applied to the biological process each day (kg BOD d}^{-1}\text{)}}{\text{mass of microorganisms in the biological process (kg biomass)}}$$

$$= \frac{\text{flow rate }(\text{m}^3\text{ d}^{-1}) \times \text{BOD }(\text{kg m}^{-3})}{\text{volume of aeration tank }(\text{m}^3) \times \text{MLSS }(\text{kg m}^{-3})}$$

MLSS is the mixed liquor suspended solids – a measure of the microbial mass and the inert substances present, determined by filtration and drying

A typical biological growth curve



In practice, biological treatment processes operate continuously, either in the high growth rate region – A – with high values of organic loading rate (F/M > 1.0), or in the low growth rate region B with low values of organic loading (F/M <0.1)

The effluent from a sedimentation tank with $200\ \mathrm{g\,m^{-3}}$ BOD, and a flow rate of 2500 $\mathrm{m^3}$ per day is to be treated biologically in a:

<u>**Low-rate process**</u> **with organic loading rate of 0.07 kg BOD per kg biomass per day used**

<u>**High-rate process**</u> **with organic loading rate of 1.8 kg BOD per kg biomass per day used**

$$\text{Mass of BOD per day} = 2500 \times 0.2 = 500\ \mathrm{kg\,d^{-1}}$$

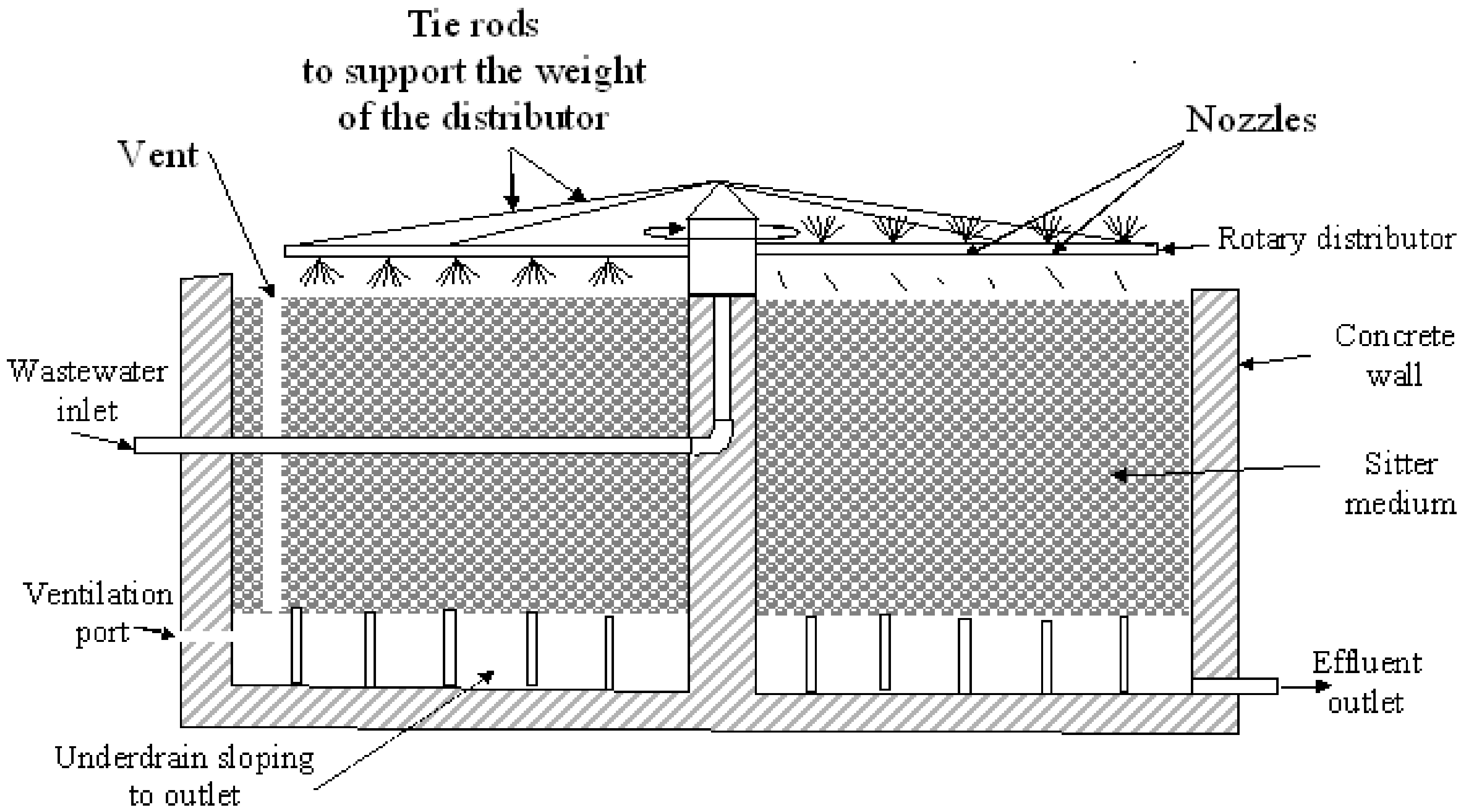
Biomass needed

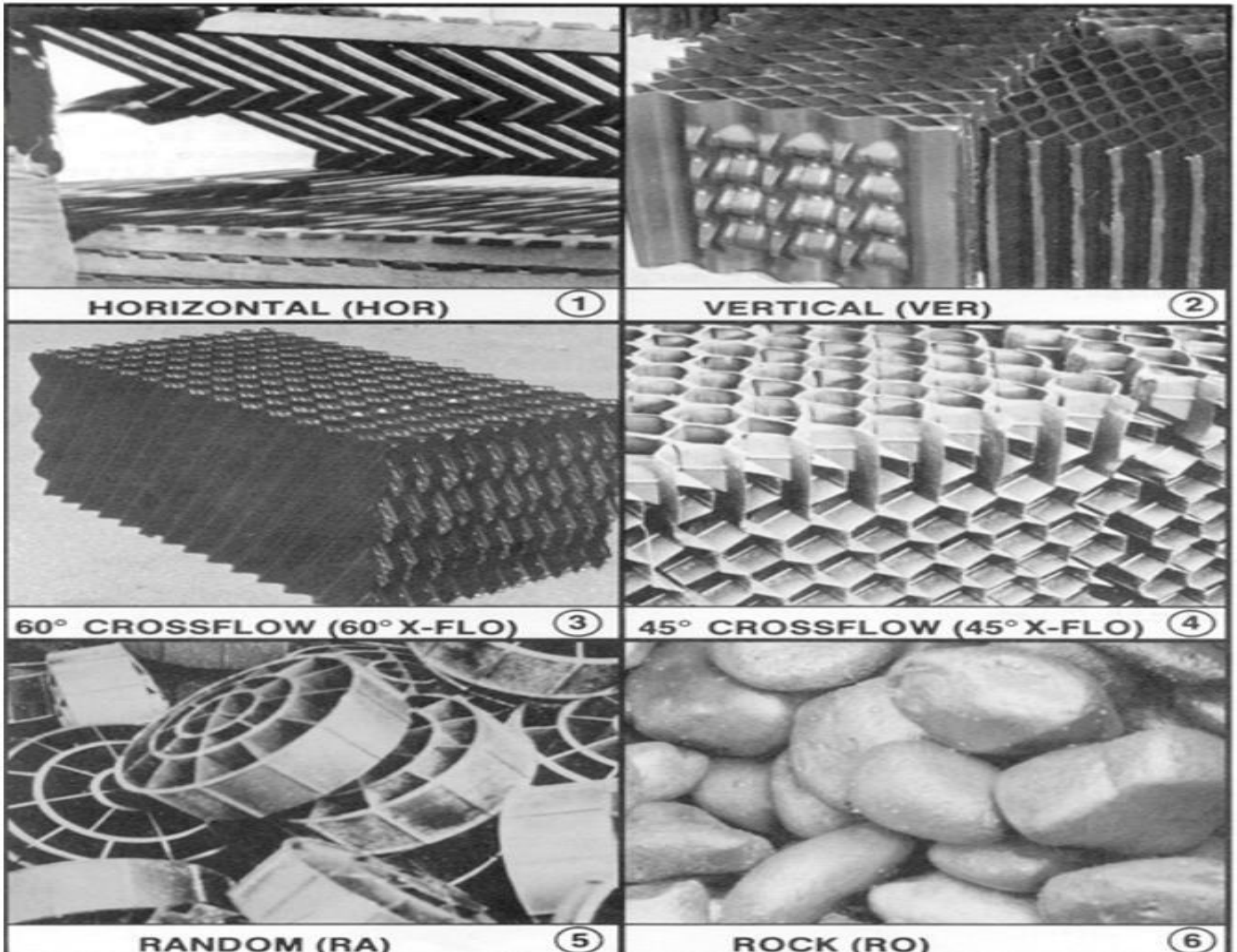
500/0.07 = 7140 kg

Biomass needed

500/1.8 = 278 kg

ATTACHED GROWTH SYSTEM :
TRICKLING FILTER SYSTEM




HORIZONTAL (HOR)
1
VERTICAL (VER)
2
60° CROSSFLOW (60° X-FLO)
3
45° CROSSFLOW (45° X-FLO)
4
RANDOM (RA)
5
ROCK (RO)
6

Low- and high-rate trickling filter processes

Filter type	Organic loading (kg BOD m^{-3} d^{-1})	Depth of filter medium (m)	BOD removal (%)
Low-rate	0.08–0.32	1.5–3.0	80–90
High-rate	0.32–1.0	1.0–2.0	50–80

Total biomass is normally calculated from the volume of the bed, and organic loading rate determined by:

$$\frac{QY}{AH} \text{ kg BOD m}^{-3} \text{ d}^{-1}$$

Q wastewater flow rate in m^3/d
A cross sectional area of filter in m^2
H bed depth in m
Y BOD of influent in kg/m^3

So many variants of biological treatment systems
e.g. Rotating Biological contactor

Activated sludge system vs. Trickling filters

Merits

- Fly nuisance avoided
- Low space requirements

Demerits

- Requires continuous attention
- Energy consumption high
- Not tolerant of variations in flow and high organic loading
- Noisy
- Operating and maintenance cost higher
- Relatively complex to monitor and control
- More vulnerable to toxic substances

Anaerobic Treatment

Used as pretreatment for High BOD waste

Why?

Organics ---→ CH_4 + energy

The conversion of organics to methane yields little energy hence the rate of growth of the microorganisms is slow and the yield of organisms by synthesis is low, therefore sludge yield is much lower than for aerobic processes. However, high process efficiency requires elevated temperature, and low BOD waste will not provide sufficient methane for heating and a supplementary source of heat will be required.

ANAEROBIC TREATMENT

- ADVANTAGES

- a considerably low sludge production
- a low mineral nutrient demand
- a lower energy input for the treatment of high-strength waste and on the other hand generation of methane gas - which may be used as a fuel for the generation of steam or electricity.
- long keeping quality of the sludge, which allows for "season operation"
- low investment and operating costs compared with other treatment methods.

- DISADVANTAGES

- Long period of time for start-up due to the slow growth rate of anaerobic bacteria
- low level of purification, which may necessitate the addition of further treatment processes to achieve a high quality

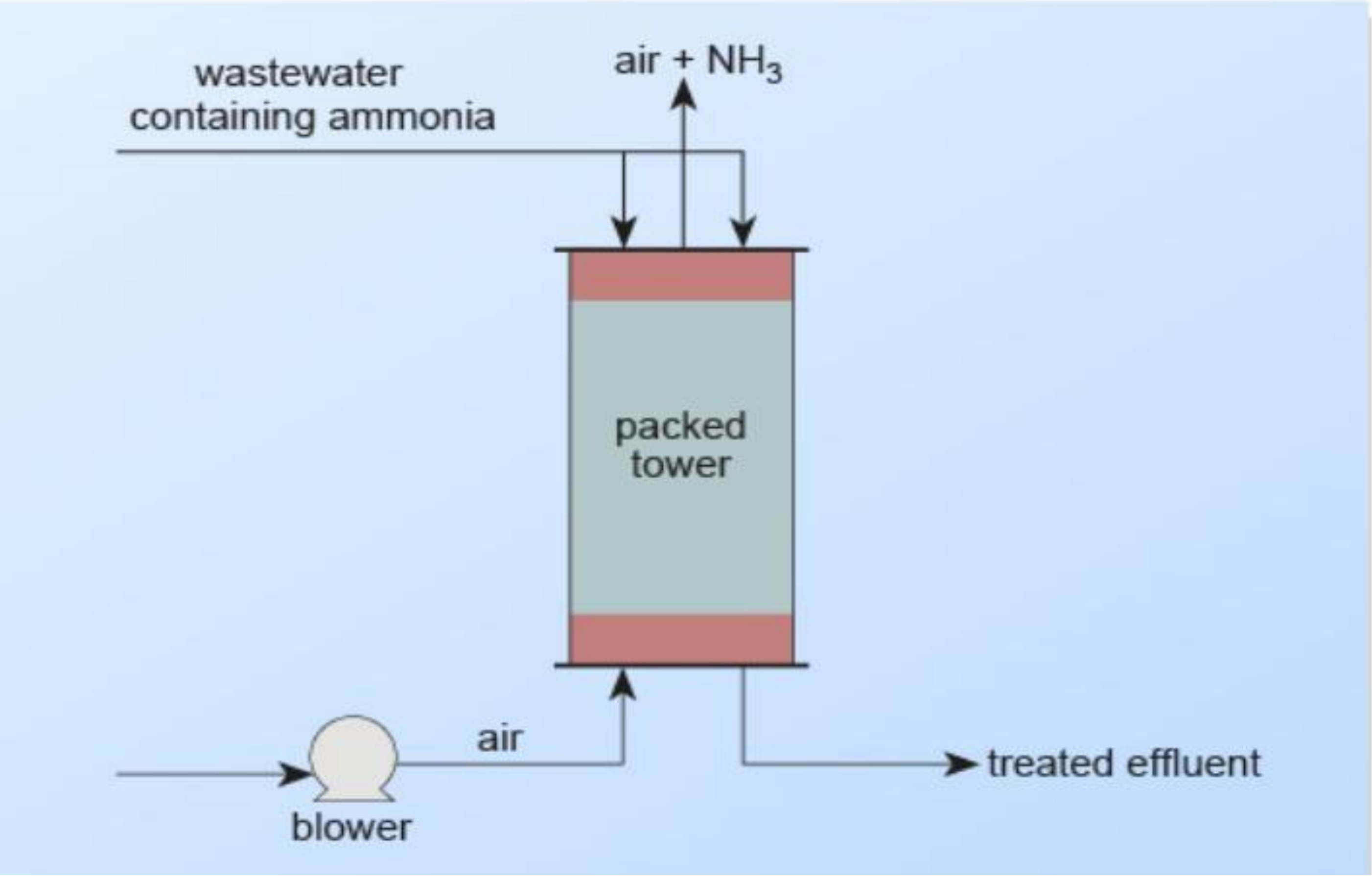
TERTIARY TREATMENT PROCESSES

Pollutants not removed in the primary and secondary treatment stage are dealt with in the tertiary treatment stage – also known as the **POLISHING STAGE** and include:

- Activated carbon (adsorption - for the removal of phosphorus and nitrogen
- Reverse osmosis
- Membrane filtration
- Ion exchange
- Disinfection / chlorination /ozonation – for bacteria and viruses and other disease causing organisms
- Coagulation, flocculation and sedimentation
- Nitrification and denitrification
- Ammonia stripping

For removing ammonia (toxic to fish)

Air stripping of ammonia



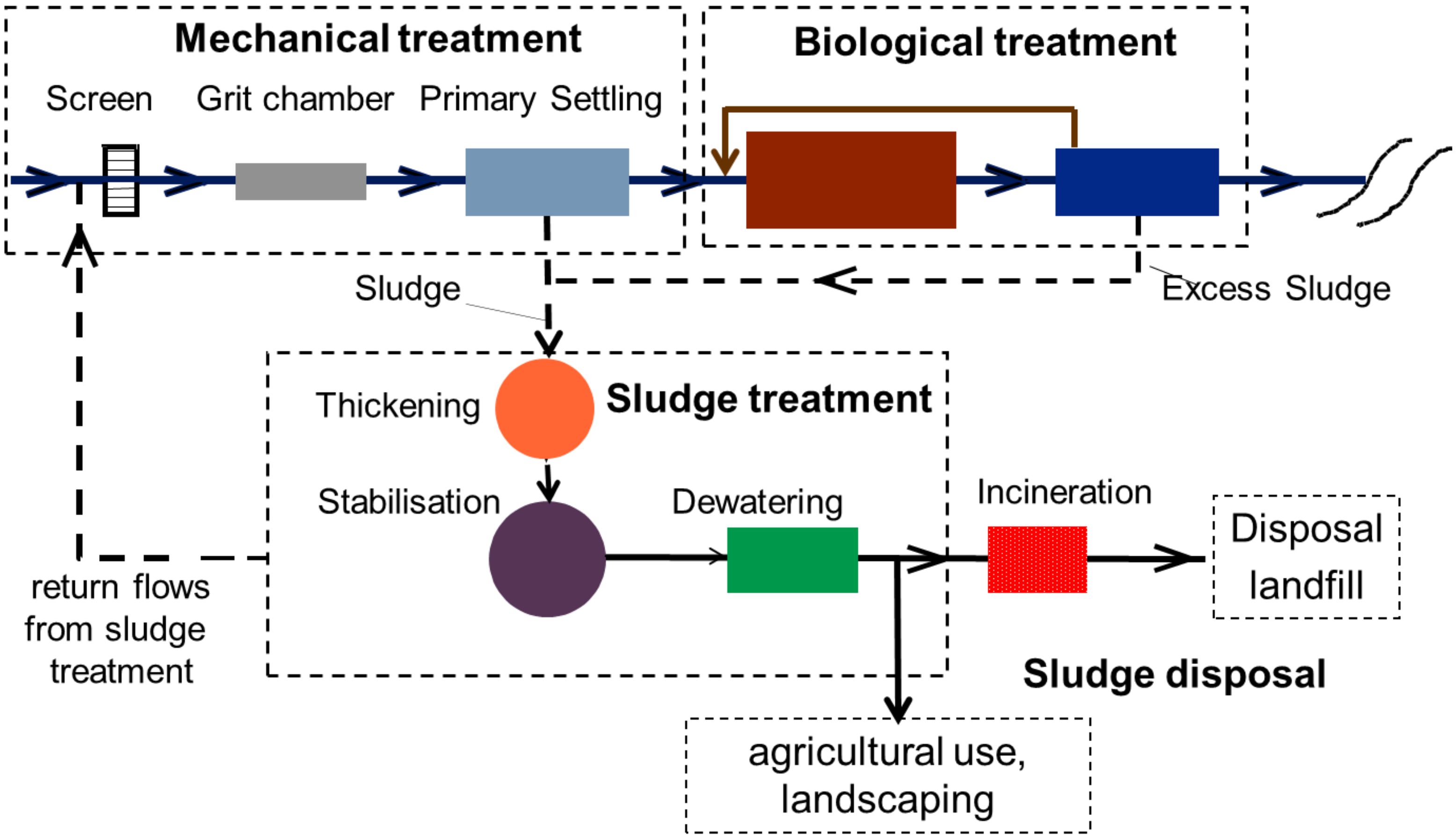
SLUDGE TREATMENT AND DISPOSAL

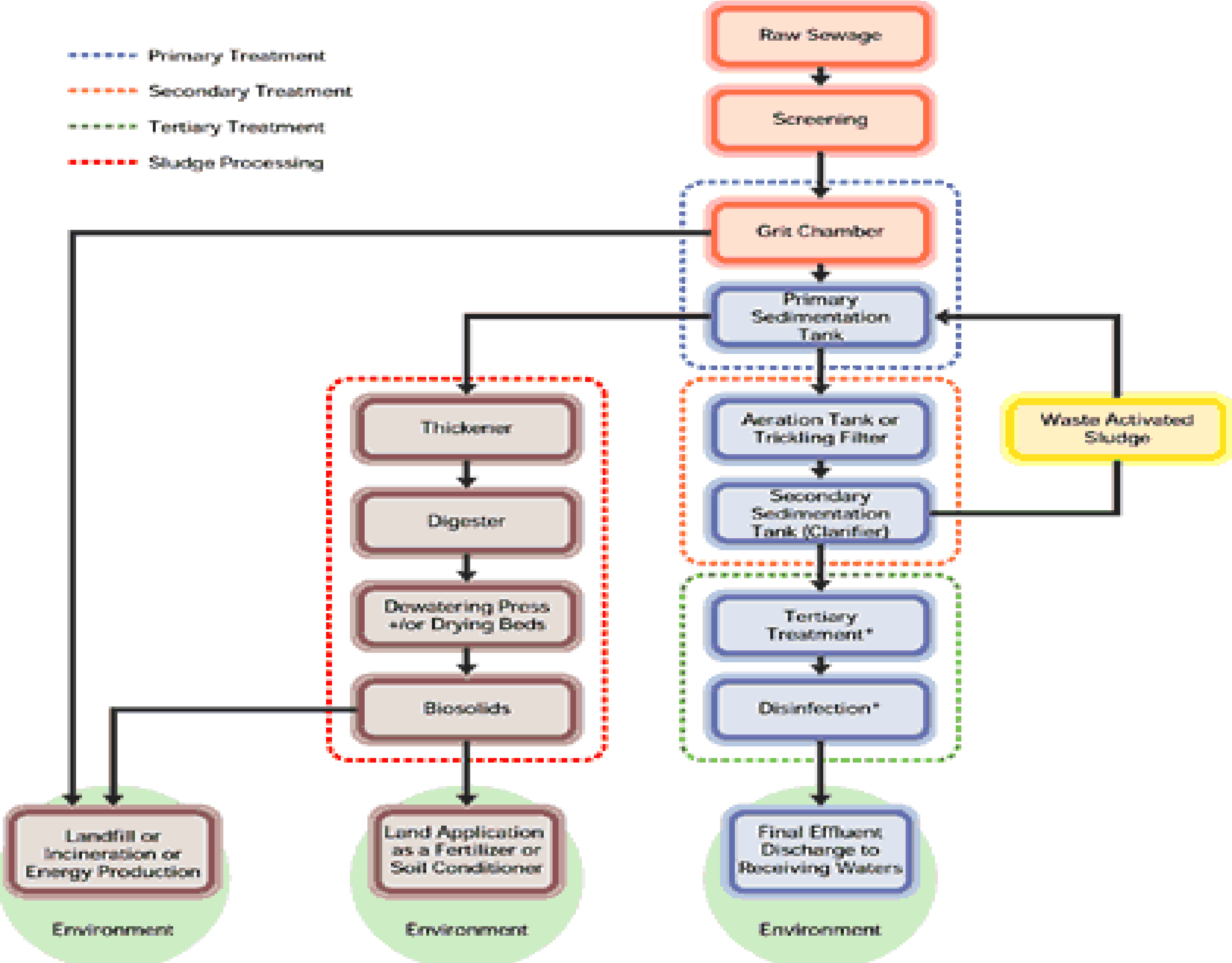
Combined sludge from the WWTP is often about 2% solids and 98% water. Because the treatment and disposal of sludge is expensive, sludge handling costs are often the overriding consideration in the design of wastewater treatment plants

Treatment processes include:

- **Thickening** - usually the first step in sludge treatment, b'cos it is impractical to handle thin sludge, a slurry of solids suspended in water. Done in sedimentation tanks or ponds
- **Anaerobic digestion** – Decomposition of organics into stable substances
- **Sludge dewatering** - Digested sludge has a lot of water, ~ 70%. This can be done on Sludge drying bed, by vacuum or pressure filtration, centrifugation, etc
- **Sludge reduction** - incineration (evaporates moisture in sludge and combusts organics to sterile ash
- **Sludge disposal** - land spreading, land filling or land disposal

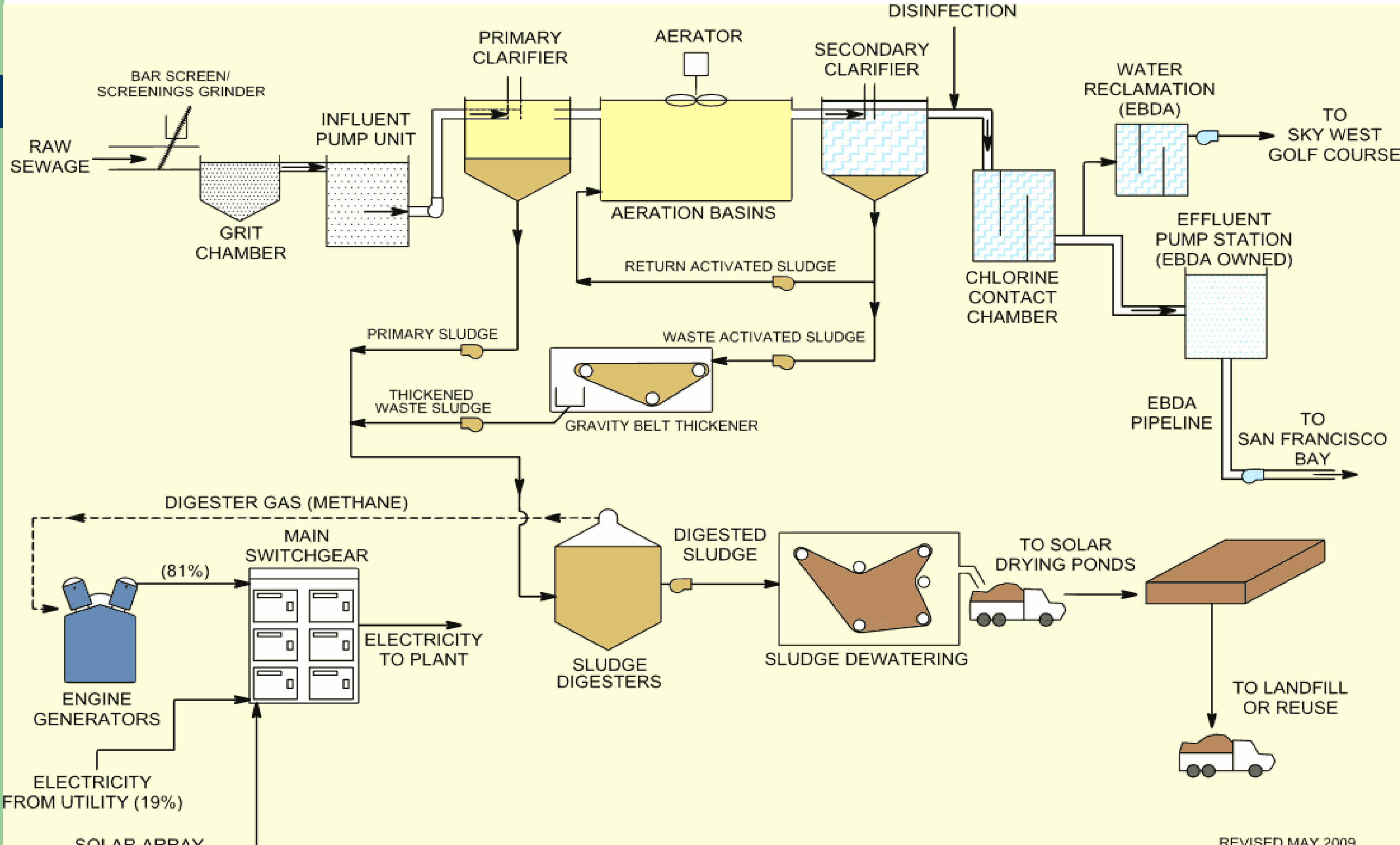
A typical WWTP showing Sludge treatment

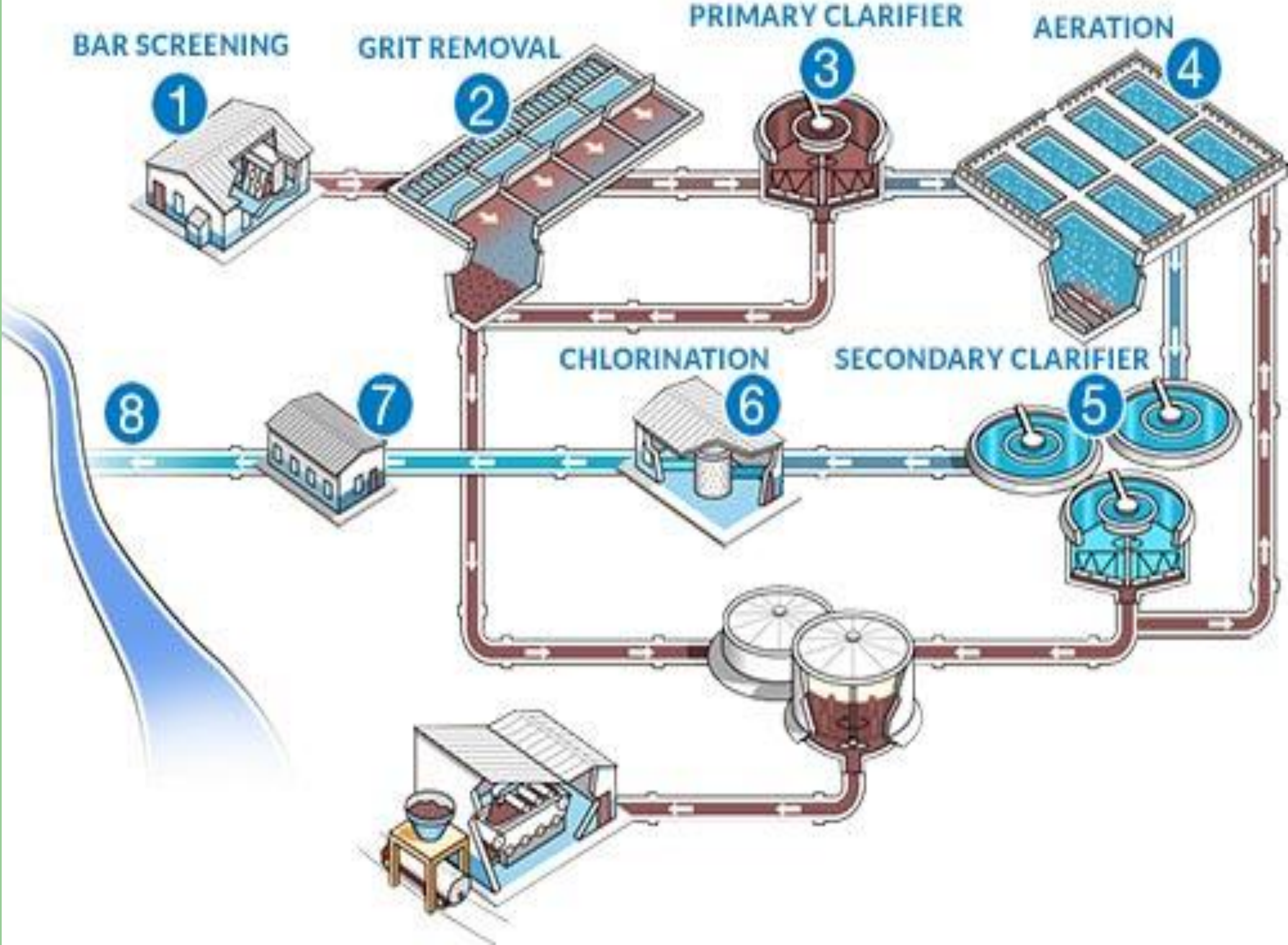




*Tertiary Treatment and Disinfection will occur only at some facilities where a very high quality effluent is required.

A typical WWTP in California (managed by East Bay Dischargers Authority (EBDA)



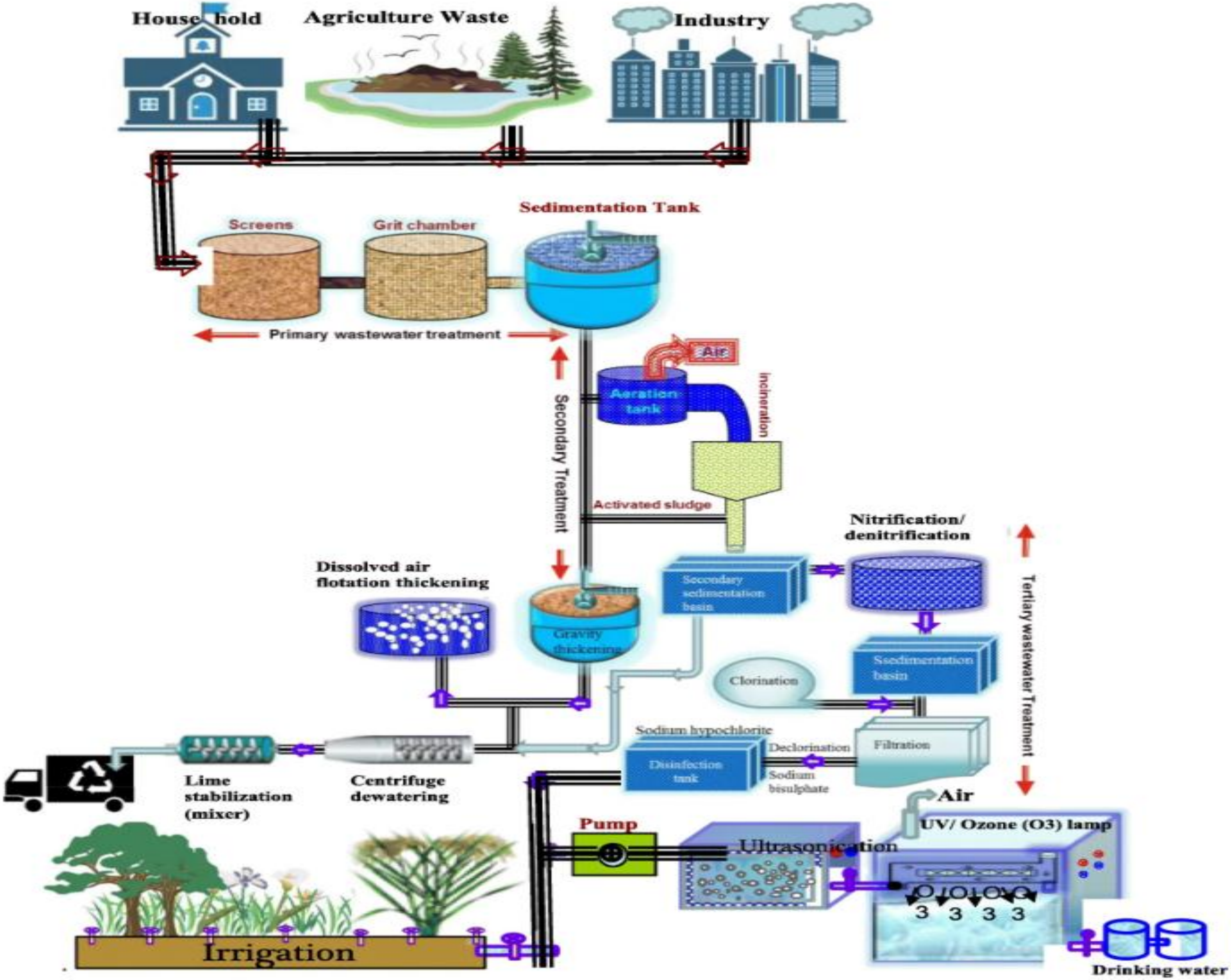

BAR SCREENING
1
GRIT REMOVAL
2
PRIMARY CLARIFIER
3
AERATION
4
CHLORINATION
6
SECONDARY CLARIFIER
5
8
7

DISPOSAL METHODS FOR TREATED WASTEWATER

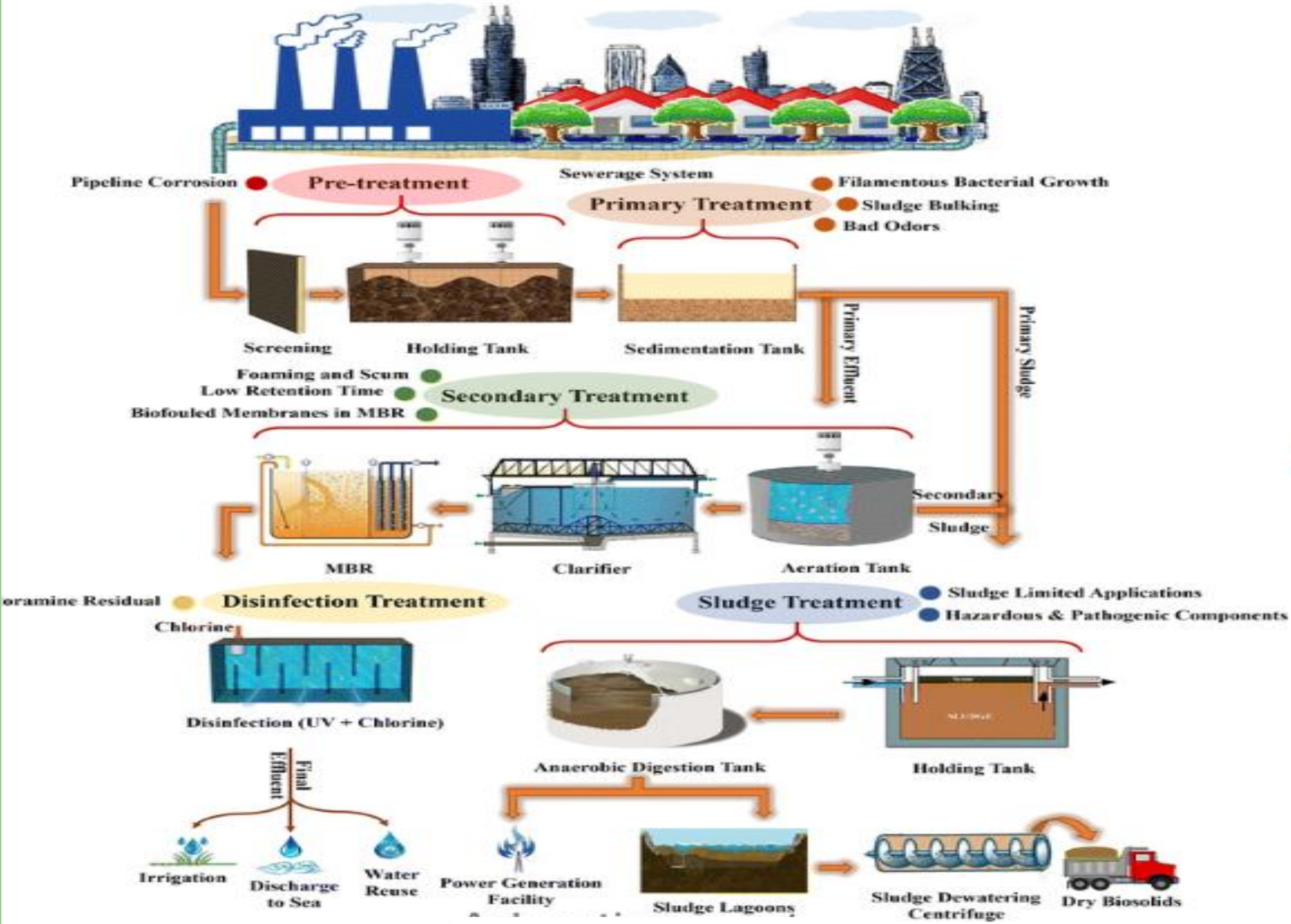
- **Natural evaporation**
- **Groundwater recharge**
- **Irrigational uses**
- **Recreational Lakes**
- **Agriculture**
- **Municipal Uses**:
- **Industrial Uses**
- **Discharge into Natural Waters**

Wasterwater injection for oil and gas recovery can cause induced seismicity (earthquakes)

A park being irrigated by treated wastewater



House hold
Agriculture Waste
Industry
Screens
Grit chamber
Sedimentation Tank
Primary wastewater treatment
Air
Aeration tank
incineration
Secondary Treatment
Activated sludge
Nitrification/ denitrification
Dissolved air flotation thickening
Gravity thickening
Secondary sedimentation basin
Tertiary wastewater Treatment
Ssedimentation basin
Clorination
Sodium hypochlorite
Declorination
Filtration
Disinfection tank
Sodium bisulphate
Lime stabilization (mixer)
Centrifuge dewatering
Air
Pump
Ultrasonication
UV/ Ozone (O3) lamp
Irrigation
Drinking water



Sewerage System
Pipeline Corrosion
Pre-treatment
Primary Treatment
Filamentous Bacterial Growth
Sludge Bulking
Bad Odors
Screening
Holding Tank
Sedimentation Tank
Primary Effluent
Primary Sludge
Foaming and Scum
Low Retention Time
Biofouled Membranes in MBR
Secondary Treatment
Secondary Sludge
MBR
Clarifier
Aeration Tank
oramine Residual
Disinfection Treatment
Chlorine
Sludge Treatment
Sludge Limited Applications
Hazardous & Pathogenic Components
Disinfection (UV + Chlorine)
Anaerobic Digestion Tank
Holding Tank
Final Effluent
Irrigation
Discharge to Sea
Water Reuse
Power Generation Facility
Sludge Lagoons
Sludge Dewatering Centrifuge
Dry Biosolids

Innovations in Industrial Wastewater Treatment

With industrial advancements come more complex pollutants – ones that traditional treatments can't deal with. So various innovations exist to

- improve existing technologies thereby making them more efficient
- Recovery energy via biogas generation
- Deal with new types of pollutants